CHAPTER 9

Acids and bases

9.1 Acids and bases ESCNZ

What are acids and bases? ESCP2

We encounter many examples of acids and bases in our daily lives. Some common examples of household items that contain acids are vinegar (contains acetic acid), lemon juice (contains citric and ascorbic acid), wine (contains tartaric acid). These acids are often found to have a sour taste. Hydrochloric acid, sulfuric acid and nitric acid are examples of acids that are more likely to be found in laboratories and industry.

Hydrochloric acid is also found in the gastric juices in the stomach. Fizzy drinks contain carbonic acid, while tea and wine contain tannic acid. People even use acids in an artistic process known as *acid etching*. In acid etching, a metal is covered in a waxy material that is resistant to acid. The bare metal is then exposed in the desired pattern and the sample is placed in an acid bath. The top layers of the exposed metal are permanently removed, creating the desired image. Some examples of this are shown below (Figure 9.2).


acetic acid (in vinegar)


citric acid (in lemons)


tartaric acid (in wine)

Figure 9.1: Some common household acids.

Figure 9.2: Some examples of the acid etching of metal.

Bases that you may know about include sodium hydroxide (commonly known as caustic soda), ammonium hydroxide and ammonia. Some of these are found in household cleaning products. Bases are usually found to have a bitter taste and feel *slippery* (soap is a good example). Acids and bases are also important commercial components in the fertiliser, plastics, and petroleum refining industries. Some common acids and bases, and their chemical formulae, are shown in Table 9.1.

Acid	Formula	Base	Formula
Hydrochloric acid	HCl	Sodium hydroxide	$NaOH$
Sulfuric acid	H_2SO_4	Potassium hydroxide	KOH
Nitric acid	HNO_3	Magnesium hydroxide	$Mg(OH)_2$
Oxalic acid	$H_2C_2O_4$	Calcium hydroxide	$Ca(OH)_2$
Sulfurous acid	H_2SO_3	Sodium bicarbonate	$NaHCO_3$
Phosphoric acid	H_3PO_4	Sodium carbonate	Na_2CO_3
Acetic (ethanoic) acid	CH_3COOH	Ammonium hydroxide	NH_4OH
Carbonic acid	H_2CO_3	Ammonia	NH_3

Table 9.1: Some common acids and bases and their chemical formulae.

FACT
The wild lupin plant takes nitrogen from the atmosphere and creates **ammonia**. It uses that ammonia to fertilise the soil for itself and the surrounding plants.

Activity: Naturally occurring acids and bases

Do research on three naturally occurring acids, and one naturally occurring base. That is, acids and bases that are found mostly in plants (not man-made).

Your research should include:

- Where the acid or base is found (e.g. the plant in which it occurs).
- Indigenous uses of the plant.
- What the chemical composition of the acid or base is.

The next section deals with some models used to describe acids and bases. It is important to have a definition so that acids and bases can be correctly identified in reactions.

FACT
Sodium bicarbonate is used in baking. It reacts with the acids in dough to release carbon dioxide. The carbon dioxide then causes the dough to expand.

Models for acids and bases ESCP3

For the acids you will encounter this year, an acid is a molecule that donates a $\mathbf{H^+}$ ion.

Substances that will act as a base include hydroxides, oxides, carbonates or hydrogen carbonate, among others. Bases often release free hydroxide ions ($\mathbf{OH^-}$) when dissociating in water.

Arrhenius model for acids and bases

A number of models for acids and bases have been developed over the years. One of the earliest was the **Arrhenius** definition. In 1884, Arrhenius discovered that water dissociates (splits up) into hydronium (H_3O^+) and hydroxide (OH^-) ions according to the following equation:

$$2H_2O(\ell) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$$

Another way of writing this is:

$$H_2O(\ell) \rightleftharpoons H^+(aq) + OH^-(aq)$$

TIP
Dissociation is the breaking apart of a molecule to form smaller molecules or ions, usually in a reversible manner. For more information on *dissociation*, refer to Grade 10.

Arrhenius described an acid as a compound that forms H_3O^+ when added to water. An Arrhenius acid therefore increases the concentration of H_3O^+ ions ($[H_3O^+]$) in water. Arrhenius described a base as a compound that dissociates in water to form OH^- ions. An Arrhenius base therefore increases the concentration of OH^- ions ($[OH^-]$) in water.

DEFINITION: *Arrhenius acids and bases*

An Arrhenius acid forms H_3O^+ in water (increases $[H_3O^+]$). An Arrhenius base forms OH^- in water (increases $[OH^-]$).

FACT
Hydrogen atoms contain only one proton. H^+ is a hydrogen atom that has lost its electron and is often called a proton.

Arrhenius acid	increases [H_3O^+]
Arrhenius base	increases [OH^-]

Table 9.2: The **Arrhenius** definition of acids and bases.

Look at the following examples showing the dissociation of hydrochloric acid and sodium hydroxide:

1. $HCl(aq) + H_2O(\ell) \rightarrow H_3O^+(aq) + Cl^-(aq)$
   Hydrochloric acid in water increases the concentration of H_3O^+ ions and is therefore an *acid*.
2. $NaOH(s) + H_2O(\ell) \rightarrow Na^+(aq) + OH^-(aq) + H_2O(\ell)$
   Sodium hydroxide in water increases the concentration of OH^- ions and is therefore a *base*.

However, this definition can only be used for acids and bases **in water**. Since there are many reactions which do not occur in water, it was important to come up with a much broader definition for acids and bases.

Brønsted-Lowry model for acids and bases

In 1923, Lowry and Brønsted took the work of Arrhenius further to develop a broader definition for acids and bases. The **Brønsted-Lowry model** defines acids and bases in terms of their ability to donate or accept protons (Table 9.3).

DEFINITION: *Brønsted-Lowry acids and bases*

An **acid** is a substance that donates (gives away) protons (H^+). A **base** is a substance that accepts (takes) protons.

Under the Brønsted-Lowry definition: an acid is a **proton donor**; a base is a **proton acceptor**.

Brønsted-Lowry acid	donates H^+	proton **donor**
Brønsted-Lowry base	accepts H^+	proton **acceptor**

Table 9.3: The **Brønsted-Lowry** definition of acids and bases.

Below are some examples:

1. $HCl(aq) + NH_3(g) \rightarrow NH_4^+(aq) + Cl^-(aq)$
   In order to decide which substance is a proton donor and which is a proton acceptor, we need to look at what happens to each reactant. The reaction can be broken down as follows:
   $HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$
   - HCl donates a proton.
     It is a *proton donor* and is therefore the **acid**.

   $NH_3(g) + H^+(aq) \rightarrow NH_4^+(aq)$
   - NH_3 accepts a proton.
     It is a *proton acceptor* and is therefore the **base**.
2. $CH_3COOH(aq) + H_2O(\ell) \rightarrow H_3O^+(aq) + CH_3COO^-(aq)$
   The reaction can be broken down as follows:
   $CH_3COOH(aq) \rightarrow CH_3COO^-(aq) + H^+(aq)$
   - CH_3COOH donates a proton.
     It is a *proton donor* and is therefore the **acid**.

$H_2O(\ell) + H^+(aq) \rightarrow H_3O^+(aq)$

- Water accepts a proton.
  It is a *proton acceptor* and is therefore the **base**.

3. $NH_3(g) + H_2O(\ell) \rightarrow NH_4^+(aq) + OH^-(aq)$
   The reaction can be broken down as follows:
   $H_2O(\ell) \rightarrow OH^-(aq) + H^+(aq)$

   - Water donates a proton.
     It is a *proton donor* and is therefore the **acid**.

   $NH_3(g) + H^+(aq) \rightarrow NH_4^+(aq)$

   - Ammonia accepts a proton.
     It is a *proton acceptor* and is therefore the **base**.

   Notice that in *example 2* **water** acted as a **base**, while in *example 3* **water** acted as an **acid**. Water can act as both an acid and a base depending on the reaction. A substance that can act as either an acid or a base is called **amphoteric**.

FACT
An amphoteric substance that contains both acidic and basic functional groups is called a **ampholyte**.

DEFINITION: *Amphoteric*

An amphoteric substance is one that can act as an acid in one reaction, or a base in another reaction.

An **amphiprotic** substance is an amphoteric substance that can donate a proton in one reaction (a Brønsted-Lowry acid), or accept a proton in another reaction (a Brønsted-Lowry base).

DEFINITION: *Amphiprotic*

An amphiprotic substance can donate a proton in one reaction, or accept a proton in another reaction.

Substances such as ammonia (NH_3), zinc oxide (ZnO), and beryllium hydroxide ($Be(OH)_2$) are amphoteric. Water and ammonia are also amphiprotic.

An acid that releases only *one* H^+ ion per molecule of acid (e.g. HCl) is referred to as **mono**protic. An acid that can release *two* H^+ ions per molecule of acid (e.g. H_2SO_4) is referred to as **di**protic. Any acid than can donate *more than one* H^+ ion per molecule of acid can be referred to as **poly**protic (this means that diprotic acids are also polyprotic).

Conjugate acid-base pairs ESCP4

Look at the reaction between hydrochloric acid and ammonia to form ammonium and chloride ions:

$HCl(aq) + NH_3(g) \leftrightharpoons NH_4^+(aq) + Cl^-(aq)$

Looking at the **forward reaction** (i.e. the reaction that proceeds from *left* to *right*), the changes that take place can be shown as follows:

- $HCl(aq) \rightarrow Cl^-(aq) + H^+(aq)$
- $NH_3(g) + H^+(aq) \rightarrow NH_4^+(aq)$

In the *forward reaction*, HCl is a proton donor (**acid**) and NH_3 is a proton acceptor (**base**).

Looking at the **reverse reaction** (i.e. the reaction that proceeds from *right* to *left*), the changes that take place are as follows:

- $Cl^-(aq) + H^+(aq) \rightarrow HCl(aq)$
- $NH_4^+(aq) \rightarrow NH_3(g) + H^+(aq)$

In the *reverse reaction*, the chloride ion (Cl^-) is the proton acceptor (**base**) and the ammonium ion (NH_4^+) is the proton donor (**acid**).

In the forward reaction HCl donates a proton (H^+) to form Cl^-. In the reverse reaction Cl^- accepts a proton to form HCl. Cl^- is the conjugate base of the acid HCl. So HCl and Cl^- are a **conjugate acid-base pair**.

Similarly, in the forward reaction NH_3 accepts a proton (H^+) to form NH_4^+. In the reverse reaction NH_4^+ donates a proton to form NH_3. NH_4^+ is the conjugate acid of the base NH_3. So NH_3 and NH_4^+ are a **conjugate acid-base pair**.

conjugate pair

$$\mathbf{HCl(aq) + NH_3(g) \rightleftharpoons NH_4^+(aq) + Cl^-(aq)}$$

acid base conjugate acid conjugate base

conjugate pair

The reaction between ammonia and water is another example:

conjugate pair

$$\mathbf{H_2O(\ell) + NH_3(g) \rightleftharpoons NH_4^+(aq) + OH^-(aq)}$$

acid base conjugate acid conjugate base

conjugate pair

DEFINITION: *Conjugate acid-base pair*

A conjugate acid-base pair contains two compounds that differ only by a hydrogen ion (H^+) and a charge of $+1$.

Worked example 1: Conjugate acid-base pairs

QUESTION

Determine the conjugate acid-base pairs for the following reaction:

$HNO_3(aq) + OH^-(aq) \rightleftharpoons NO_3^-(aq) + H_2O(\ell)$

SOLUTION

Step 1: Which reactant is an acid and which is a base?

HNO_3 is nitric acid. It donates a proton in the forward reaction:

$HNO_3(aq) \rightarrow NO_3^-(aq) + H^+(aq)$

OH^- accepts a proton in the forward reaction and is therefore the base:

$OH^-(aq) + H^+(aq) \rightarrow H_2O(\ell)$

Step 2: Which product is the conjugate base of the acid?

Nitric acid (HNO_3) donates a proton to become NO_3^-. In the reverse reaction NO_3^- accepts a proton to become HNO_3. Therefore NO_3^- is the conjugate base of HNO_3.

$NO_3^-(aq) + H^+(aq) \rightarrow HNO_3(aq)$

Step 3: Which product is the conjugate acid of the base?

OH^- accepts a proton to become H_2O. In the reverse reaction H_2O donates a proton to become OH^-. Therefore H_2O is the conjugate acid of OH^-.

$H_2O(\ell) \rightarrow OH^-(aq) + H^+(aq)$

Step 4: Label the conjugate acid-base pairs in this reaction

conjugate pair

$$\underset{\text{acid}}{HNO_3(aq)} + \underset{\text{base}}{OH^-(aq)} \rightleftharpoons \underset{\text{conjugate base}}{NO_3^-(aq)} + \underset{\text{conjugate acid}}{H_2O(\ell)}$$

conjugate pair

Exercise 9 – 1: Acids and bases

1. In each of the following reactions, label the conjugate acid-base pairs.
   a) $H_2SO_4(aq) + H_2O(aq) \rightleftharpoons H_3O^+(aq) + HSO_4^-(aq)$
   b) $NH_4^+(aq) + F^-(aq) \rightleftharpoons HF(aq) + NH_3(g)$
   c) $H_2O(\ell) + CH_3COO^-(aq) \rightleftharpoons OH^-(aq) + CH_3COOH(aq)$
   d) $H_2SO_4(aq) + Cl^-(aq) \rightleftharpoons HCl(aq) + HSO_4^-(aq)$
2. Write the conjugate base for:
   a) H_2CO_3
   b) H_3O^+
   c) $H_2PO_4^-$
3. Write the conjugate acid for:
   a) OH^-
   b) CN^-
   c) CO_3^{2-}
4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27W3 2a. 27W4 2b. 27W5 2c. 27W6 3a. 27W7 3b. 27W8 3c. 27W9

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Strong and weak acids and bases ESCP5

Strong acids and bases

A strong acid or base is one that will almost completely *dissociate* or *ionise* to form ions in solution. That is, a large percentage of the moles of a strong acid or base will form ions when added to water.

DEFINITION: *Strong acid and strong base*

A **strong** acid or base is one that almost completely dissociates to form ions in solution.

HCl is a strong acid. For example if 100 000 molecules of HCl are added to water and 90 000 ionise to form H^+ and Cl^- ions, then there is a large amount of ionisation. This is what makes HCl a strong acid.

$$\mathbf{HCl(g) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + Cl^-(aq)}$$

almost complete ionisation

large percentage of molecules ionised

The unequal double arrows in the reaction equation indicate that the equilibrium position favours the formation of ions.

Thee are three strong acids that we commonly find: HCl (hydrochloric acid), HNO_3 (nitric acid) and H_2SO_4 (sulfuric acid). Two strong bases that are commonly found are: NaOH (sodium hydroxide) and KOH (potassium hydroxide).

Weak acids and bases

A weak acid or base is one where only a *small percentage* of molecules will dissociate to form ions in solution.

DEFINITION: *Weak acid and weak base*

A **weak** acid or base is one where only a small percentage of molecules dissociate to form ions in solution.

HF is a weak acid. For example if 100 000 molecules of HF are added to water and only 100 ionise to form H^+ and F^- ions, then there is only a small amount of ionisation. This is what makes HF a weak acid.

$$\mathbf{HF(g) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + F^-(aq)}$$

very little ionisation

small percentage of molecules ionised

The unequal double arrows in the reaction equation indicate that the equilibrium position does not favour the formation of ions.

An example of a weak base is $Mg(OH)_2$ which will only dissociate partially into Mg^{2+} and OH^- ions.

Dilute and concentrated solutions ESCP6

A different concept to strong and weak is the concept of **concentrated** and **dilute**. Where strong and weak refer to the characteristic of a compound, concentrated and dilute refer to the characteristic of a solution. Thus a strong acid can be prepared as either a concentrated or a dilute solution. A solution of which the exact concentration is known is called a **standard solution**.

DEFINITION: *Standard solution*

A standard solution is one where the exact concentration of solute in a solvent is known.

Concentrated solutions

A concentrated solution is one where a large amount of a substance (solute) has been added to a solvent. Note that both strong and weak acids and bases can be used in concentrated solutions.

DEFINITION: *Concentrated solution*

A concentrated solution is one where there is a high ratio of dissolved substance (e.g. acid or base) to solvent.

Dilute solutions

A dilute solution is one where a small amount of a substance has been added to a solvent. Note that both strong and weak acids and bases can be used in dilute solutions.

DEFINITION: *Dilute solution*

A dilute solution is one where there is a low ratio of dissolved substance to solvent.

A concentrated solution can be made from a strong or a weak acid or base. A dilute solution can also be made from a strong or a weak acid or base. Whether a solution is concentrated or dilute depends on how much of the acid or base was added to the solvent.

A **strong base** that is also **concentrated** would be a base that **almost completely dissociates** when added to a solution, and you also add a **large amount** of the base to the solution.

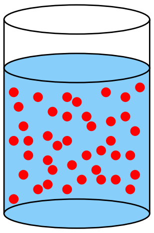
A **weak acid** that is also **dilute** would be an acid where only a **small percentage of molecules ionise** when added to a solution, and you also add only a **small amount** of the acid to the solution. Table 9.4 summarises these concepts.

The electrical conductivity of a solution depends on the concentration of **mobile ions** in the solution. This means that a concentrated solution of a strong acid or base will have a high electrical conductivity, while a dilute solution of a weak acid or base will have a low electrical conductivity.

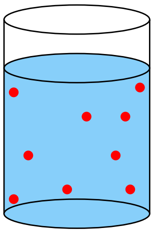
	Acid	Base
Strong	**high percentage** forms ions in solution	**high percentage** forms ions in solution
Weak	only a **small percentage** forms ions in solution	only a **small percentage** forms ions in solution
Concentrated	**large number** of moles of acid in solution	**large number** of moles of base in solution
Dilute	**small number** of moles of acid in solution	**small number** of moles of base in solution

Table 9.4: A summary of the properties of strong, weak, concentrated, and dilute acids and bases.

TIP



A concentrated solution has a lot of solute molecules (red circles) in the solvent.



A dilute solution has few solute molecules (red circles) in the solvent.

FACT
Electric current is the movement of charged particles. Therefore, the more ions (charged particles) there are in a solution, the greater the electric current that can be conducted through the solution by the charged particles. This is the electrical conductivity of a solution.

TIP
Mobile ions are ions that are able to move. These include ions in solution and ions in melted ionic materials. The ions in ionic solids are not mobile.

Worked example 2: Acids and bases

QUESTION

Solution 1 contains 100 dm^3 of HCl added to 10 dm^3 of water. Almost all the HCl molecules ionise in the solution.

Solution 2 contains 0,01 g of $Mg(OH)_2$ added to 1000 dm^3 of water. Only a small percentage of the $Mg(OH)_2$ molecules dissociate in the solution.

Say whether these solutions:

1. Contain a strong or weak acid or base.
2. Are concentrated or dilute.

SOLUTION

Step 1: Are the compounds acids or bases?

HCl is hydrochloric acid. It would donate a proton and is an acid. $Mg(OH)_2$ is magnesium hydroxide and is a base.

Step 2: What makes an acid or base strong or weak?

Almost complete ionisation or dissociation means an acid or base is strong. Only a small amount of ionisation or dissociation means an acid or base is weak.

Step 3: Are the compounds strong or weak acids and bases?

Almost all the HCl molecules ionise in the solution, therefore HCl is a strong acid.

Only a small percentage of the $Mg(OH)_2$ molecules dissociate, therefore $Mg(OH)_2$ is a weak base.

Step 4: What makes a solution concentrated or dilute?

A concentrated solution has a high ratio of solute to solvent. A dilute solution has a low ratio of solute to solvent.

Step 5: Are the solutions concentrated or dilute?

100 dm^3 of HCl is added to 10 dm^3 of water. This is a high ratio, therefore the solution of HCl is concentrated.

0,01 g of $Mg(OH)_2$ is added to 1000 dm^3 of water. This is a low ratio, therefore the solution of $Mg(OH)_2$ is dilute.

Step 6: Combine your information

Solution 1 is a **concentrated solution** of a **strong acid**.

Solution 2 is a **dilute solution** of a **weak base**.

Worked example 3: Acids and bases

QUESTION

Solution 1 contains 0,01 dm^3 of NaOH added to 800 dm^3 of water. Almost all the NaOH molecules dissociate in the solution.

Solution 2 contains 100 g of HF added to 10 dm^3 of water. Only a small percentage of the HF molecules ionise in the solution.

Say whether these solutions:

1. Contain a strong or weak acid or base.
2. Are concentrated or dilute.

SOLUTION

Step 1: Are the compounds acids or bases?

$NaOH$ is sodium hydroxide and is a base. HF is hydrofluoric acid and is an acid.

Step 2: What makes an acid or base strong or weak?

Almost complete ionisation or dissociation means an acid or base is strong. Only a small amount of ionisation or dissociation means an acid or base is weak.

Step 3: Are the compounds strong or weak acids and bases?

Almost all the $NaOH$ molecules dissociate in the solution, therefore $NaOH$ is a strong base.

Only a small percentage of the HF molecules ionise, therefore HF is a weak acid.

Step 4: What makes a solution concentrated or dilute?

A concentrated solution has a high ratio of solute to solvent. A dilute solution has a low ratio of solute to solvent.

Step 5: Are the solutions concentrated or dilute?

0,01 g of $NaOH$ is added to 800 dm^3 of water. This is a low ratio, therefore the solution of $NaOH$ is dilute.

100 dm^3 of HF is added to 10 dm^3 of water. This is a high ratio, therefore the solution of HF is concentrated.

Step 6: Combine your information

Solution 1 is a **dilute solution** of a **strong base**.

Solution 2 is a **concentrated solution** of a **weak acid**.

TIP

Remember, to calculate the concentration of a solution we use the formula:

$$C = \frac{n}{V}$$

Worked example 4: Calculating concentration

QUESTION

0,27 g of H_2SO_4 is added to 183,7 dm^3 of water. Calculate the concentration of the solution.

SOLUTION

Step 1: List the information you have and the information you need

$V = 183{,}7\ dm^3$, $m = 0{,}27\ g$

The volume (V) and the mass (m) are given. The number of moles (n) needs to be calculated. To do that the molar mass (M) needs to be calculated.

Step 2: Make sure all given units are correct and convert them if necessary

All the units are correct.

Step 3: What equations will be necessary to calculate the concentration?

$$C\ (mol.dm^{-3}) = \frac{n\ (mol)}{V\ (dm^3)} \qquad n\ (mol) = \frac{m\ (g)}{M\ (g.mol^{-1})}$$

Step 4: Calculate the number of moles of the acid in the solution

$M(H_2SO_4) = (2 \times 1{,}01) + 32{,}1 + (4 \times 16) = 98{,}12\ g.mol^{-1}$.

$$n = \frac{m}{M} = \frac{0{,}27\ g}{98{,}12\ g.mol^{-1}} = 0{,}0028\ mol$$

Step 5: Calculate the concentration of the solution

$$C = \frac{n}{V} = \frac{0,0028 \text{ mol}}{183,7 \text{ dm}^3} = 0,0000152 \text{ mol.dm}^{-3} = 1,52 \times 10^{-5} \text{ mol.dm}^{-3}$$

Worked example 5: Calculating concentration

QUESTION

16,4 g of KOH is added to 12,9 cm^3 of water. Calculate the concentration of the solution.

SOLUTION

Step 1: List the information you have and the information you need

$V = 12,9 \text{ cm}^3$, $m = 16,4 \text{ g}$

The volume (V) and the mass (m) are given. The number of moles (n) needs to be calculated. To do that the molar mass (M) needs to be calculated.

Step 2: Make sure all given units are correct or convert them

The volume needs to be converted to dm^3.

$$V = 12,9 \text{ cm}^3 \times \frac{0,001 \text{ dm}^3}{1 \text{ cm}^3} = 0,0129 \text{ dm}^3$$

Step 3: What equations will be necessary to calculate the concentration?

$$C \text{ (mol.dm}^{-3}) = \frac{n \text{ (mol)}}{V \text{ (dm}^3)} \qquad n \text{ (mol)} = \frac{m \text{ (g)}}{M \text{ (g.mol}^{-1})}$$

Step 4: Calculate the number of moles of base in the solution

$M(KOH) = 39,1 + 16 + 1,01 = 56,11 \text{ g.mol}^{-1}$.

$$n = \frac{m}{M} = \frac{16,4 \text{ g}}{56,11 \text{ g.mol}^{-1}} = 0,292 \text{ mol}$$

Step 5: Calculate the concentration of the solution

$$C = \frac{n}{V} = \frac{0,292 \text{ mol}}{0,0129 \text{ dm}^3} = 22,64 \text{ mol.dm}^{-3}$$

Exercise 9 – 2: Types of acids and bases

1. Say whether the solutions of the acids and bases in the following situations are concentrated or dilute.
   a) For every 1 mole of a solvent there are 50 moles of lithium hydroxide (LiOH).
   b) For every 100 moles of a solvent there are 5 moles of nitric acid (HNO_3).
2. 95% of an unknown acid donates protons when the acid is added to water. The pH of the final solution is 6,5.
   a) Is the acid a strong or a weak acid? Give a reason for your answer.
   b) Is a solution with a pH of 6,5 strongly or weakly acidic?
   c) Is the solution concentrated or dilute? Give a reason for your answer.

3. Calculate the concentration for of the following solutions.
   a) 27 g of sodium bicarbonate ($NaHCO_3$) added to 22,6 cm^3 of water.
   b) 0,893 mol of phosphoric acid (H_3PO_4) added to 4,79 dm^3 of a solvent.
   c) 32,8 mg of hydrochloric acid (HCl) added to 12,76 cm^3 of water.
   d) 1,12 dm^3 of a 6,54 $mol.dm^{-3}$ concentration solution of ammonia (NH_3) added to 0,50 dm^3 of water.
4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 27WB 1b. 27WC 2. 27WD 3a. 27WF 3b. 27WG 3c. 27WH
3d. 27WJ

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K_a and K_b ESCP7

The equilibrium constant for the ionisation process of an acid (the extent to which ions are formed in solution) is given by the term K_a, while that for a base is given by K_b. These equilibrium constants are a way of determining whether the acid or base is weak or strong.

Remember from Chapter 8 that we calculate K as follows: $K = \frac{[\text{products}]}{[\text{reactants}]}$

Acids

- **Strong acids**

  Consider the ionisation of HBr:

  $HBr(g) + H_2O(\ell) \rightarrow H_3O^+(aq) + Br^-(aq)$

  We are calculating K_a here as it is an acid being dissolved in water. The *liquid* water is not included in the equation.

$$K_a = \frac{[H_3O^+(aq)][Br^-(aq)]}{[HBr(g)]}$$

  The value of K_a for this reaction is very high (approximately 1×10^9). This means that there are many more moles of product than reactant. The HBr molecules are almost all ionised to H^+ and Br^-.

  We can then show that in the ionisation reaction HBr is almost completely ionised. Effectively, any value above 1×10^3 is large and shows that almost complete ionisation has occurred:

$$\mathbf{HBr(g) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + Br^-(aq)}$$

almost complete ionisation

large percentage of molecules ionised

  The unequal double arrows in the reaction equation indicate that the equilibrium position favours the formation of ions. This is in contrast to the ionisation of a weak acid.

FACT
There are really only six strong inorganic acids, the rest are considered weak. These are $HClO_4$ (perchloric acid), HI (hydroiodic acid), HBr (hydrobromic acid), HCl (hydrochloric acid), H_2SO_4 (sulfuric acid) and HNO_3 (nitric acid).

- **Weak acids**

  Consider the ionisation of ethanoic acid (CH_3COOH):

  $CH_3COOH(aq) + H_2O(\ell) \rightarrow H_3O^+(aq) + CH_3COO^-(aq)$

$$K_a = \frac{[H_3O^+(aq)][CH_3COO^-(aq)]}{[CH_3COOH(aq)]}$$

  The value of K_a for this reaction is very low (approximately $1{,}7 \times 10^{-5}$), showing that only a few of the CH_3COOH molecules ionise to form H_3O^+ (in water) and CH_3COO^-.

  We can write the reaction with an unequal double arrow to show the position of the equilibrium, which does not favour the formation of ions:

$$\mathbf{CH_3COOH(aq) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + CH_3COO^-(aq)}$$

very little ionisation

small percentage of molecules ionised

Name	Formula	K_a values	Type
Hydrobromic acid	HBr	$1{,}0 \times 10^9$	strong acid
Hydrochloric acid	HCl	$1{,}3 \times 10^6$	strong acid
Sulfuric acid	H_2SO_4	**First** H^+: $1{,}0 \times 10^3$ **Second** H^+: $1{,}0 \times 10^{-2}$	strong acid
Oxalic acid	$H_2C_2O_4$	**First** H^+: $5{,}8 \times 10^{-2}$ **Second** H^+: $6{,}5 \times 10^{-5}$	weak acid
Sulfurous acid	H_2SO_3	**First** H^+: $1{,}4 \times 10^{-2}$ **Second** H^+: $6{,}3 \times 10^{-8}$	weak acid
Hydrofluoric acid	HF	$3{,}5 \times 10^{-4}$	weak acid
Ethanoic acid	CH_3COOH	$1{,}7 \times 10^{-5}$	weak acid

Table 9.5: K_a values for the ionisation of some common acids. Note that sulfuric, oxalic and sulfurous acid can all lose 2 H^+ ions (they are diprotic acids).

Bases

The dissociation of bases is similar to that of acids in that we look at the K_b values in a similar manner:

- **Strong bases**

  For a strong base like NaOH:

  $NaOH(aq) + H_2O(\ell) \rightarrow Na^+(aq) + OH^-(aq) + H_2O(\ell)$

$$K_b = \frac{[Na^+(aq)][OH^-(aq)]}{[NaOH(aq)]}$$

  The K_b for NaOH is very large (NaOH is a strong base and almost completely dissociates) and shows that the equilibrium lies very far to the OH^- side of the reaction. As a result we can write the equilibrium as:

$$\mathbf{NaOH_{(aq)} \rightleftharpoons Na^+{}_{(aq)} + OH^-{}_{(aq)}}$$

almost complete dissociation

large percentage of molecules dissociate

- **Weak bases**

  $NH_3(g) + H_2O(\ell) \rightarrow NH_4^+(aq) + OH^-(aq)$

$$K_b = \frac{[NH_4^+(aq)][OH^-(aq)]}{[NH_3(g)]}$$

The K_b for NH_3 is approximately $1,8 \times 10^{-5}$ (NH_3 is a weak base) and shows that the equilibrium lies to the NH_3 side of the reaction. As a result we can write the equilibrium as:

$$NH_3(g) + H_2O(\ell) \rightleftharpoons NH_4{}^+(aq) + OH^-(aq)$$

very little ionisation
small percentage of molecules ionised

TIP
In a balanced chemical equation such as:
$2C \rightarrow A + B$
$K_a = \frac{[A]^1[B]^1}{[C]^2}$
However, we generally do not include the superscripts when the coefficients in the balanced equation are **1**:
$K_a = \frac{[A][B]}{[C]^2}$
Remember: $x^1 = x$.

Worked example 6: Equilibrium constant calculations

QUESTION

Calculate the equilibrium constant for hydrochloric acid added to $1,38\ dm^3$ of water:

$HCl(aq) + H_2O(\ell) \rightarrow H_3O^+(aq) + Cl^-(aq)$

n(HCl) in solution = 0,005 mol

n(Cl^-) in solution = 87,3 mol

SOLUTION

Step 1: Calculate the concentration of HCl at equilibrium

$$C(HCl) = \frac{n}{V} = \frac{0,005\ mol}{1,38\ dm^3} = 0,0036\ mol.dm^{-3}$$

Step 2: Calculate the concentration of Cl^- at equilibrium

$$C(Cl^-) = \frac{n}{V} = \frac{87,3\ mol}{1,38\ dm^3} = 63,3\ mol.dm^{-3}$$

Step 3: Calculate the concentration of H_3O^+ at equilibrium

There is a 1:1 mole ratio between H_3O^+ and Cl^-. Therefore, if 87,3 mol of Cl^- is present at equilibrium, 87,3 mol of H_3O^+ must be present as well.

$$C(H_3O^+) = \frac{n}{V} = \frac{87,3\ mol}{1,38\ dm^3} = 63,3\ mol.dm^{-3}$$

Step 4: Calculate K_a (because HCl is an acid)

$$K_a = \frac{[H_3O^+(aq)]^1[Cl^-(aq)]^1}{[HCl]^1}$$

$$K_a = \frac{63,3 \times 63,3}{0,0036} = 1,11 \times 10^6$$

Worked example 7: Equilibrium constant calculations

QUESTION

The equilibrium constant (K_b) for the following reaction is $1,8 \times 10^{-5}$.

$NH_3(aq) + H_2O(\ell) \rightarrow NH_4^+(aq) + OH^-(aq)$

Calculate the mass (at equilibrium) of NH_3 molecules dissolved in $4\ dm^3$ of water if there is a $0,00175\ mol.dm^{-3}$ concentration of hydroxide ions at equilibrium.

SOLUTION

Step 1: What is the K_b equation for this reaction

$$K_b = \frac{[NH_4^+(aq)]^1[OH^-(aq)]^1}{[NH_3(aq)]^1}$$

Step 2: Calculate the concentration of NH_3 at equilibrium

If there is a 0,00175 $mol.dm^{-3}$ concentration of OH^- ions at equilibrium there must be an equal concentration of NH_4^+ ions.

$$[NH_3(aq)] = \frac{[NH_4^+(aq)]^1[OH^-(aq)]^1}{K_b} = \frac{(0,00175)(0,00175)}{1,8 \times 10^{-5}} = 0,17\ mol.dm^{-3}$$

Step 3: Calculate the number of moles of NH_3 at equilibrium

$$C\ (mol.dm^{-3}) = \frac{n\ (mol)}{V\ (dm^3)},\ \text{therefore:}$$

$$n = C \times V = 0,17\ mol.dm^{-3} \times 4\ dm^3 = 0,68\ mol$$

Step 4: Calculate the mass of NH_3 in solution at equilibrium

$$n\ (mol) = \frac{m\ (g)}{M\ (g.mol^{-1})}.\ \text{Therefore } m = n \times M$$

$$M(NH_3) = 14,0 + (3 \times 1,01) = 17,03\ g.mol^{-1}$$

$$m = 0,68\ mol \times 17,03\ g.mol^{-1} = 11,6\ g$$

Exercise 9 – 3: Acids and bases

1. State whether the acids and bases in the following balanced chemical equations are strong or weak and give a reason for your answer.
   a) HF: $\mathbf{HF(g) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + F^-(aq)}$
   b) KOH dissolved in water: $\mathbf{KOH(s) \rightleftharpoons K^+(aq) + OH^-(aq)}$
   c) NH_3: $\mathbf{NH_3(g) + H_2O(\ell) \rightleftharpoons NH_4{}^+(aq) + OH^-(aq)}$
   d) HCl $\mathbf{HCl(g) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + Cl^-(aq)}$
2. State whether the acids and bases in the following questions are strong or weak and give a reason for your answer.
   a) CH_3COOH has a K_a value of $1,7 \times 10^{-5}$.
   b) NH_3 has a K_b value of $1,8 \times 10^{-5}$.
   c) HI has a K_a value of $3,2 \times 10^9$.
3. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 27WK 1b. 27WM 1c. 27WN 1d. 27WP 2a. 27WQ 2b. 27WR
2c. 27WS

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9.2 Acid-base reactions ESCP8

When an acid and a base react, they form a **salt**. If the base contains hydroxide (OH^-) ions, then *water* will also be formed. The word *salt* is a general term which applies to the products of all acid-base reactions. A salt is a product that is made up of the **cation** from a **base** and the **anion** from an **acid**.

DEFINITION: *Salt*

A salt is a neutral ionic compound composed of cations and anions. It is the result of an acid-base neutralisation reaction.

Neutralisation reactions ESCP9

When an equivalent amount of acid and base react (so that neither the acid nor the base are in excess), the reaction is said to have reached the **equivalence** point. At this point **neutralisation** has been achieved.

DEFINITION: *Equivalence point*

When a stoichiometrically equivalent number of moles of both reactants has been added to the reaction vessel.

To better understand stoichiometric equivalence look at the following equations:

1. $1HA(aq) + 1BOH(aq) \rightarrow 1AB(aq) + 1H_2O(\ell)$
2. $1H_2A(aq) + 2BOH(aq) \rightarrow 1AB_2(aq) + 2H_2O(\ell)$

In the first example above, a stoichiometrically equivalent number of moles is **one** mole of HA for every **one** mole of BOH. In the second example, a stoichiometrically equivalent number of moles is **one** mole of H_2A for every **two** moles of BOH.

DEFINITION: *Neutralisation*

A neutralisation reaction involves an acid and a base reacting to form a salt.

Look at the following examples:

- **Hydrochloric acid with sodium hydroxide**
  Hydrochloric acid reacts with sodium hydroxide to form sodium chloride (the salt) and water. Sodium chloride is made up of Na^+ cations from the base (NaOH) and Cl^- anions from the acid (HCl).
  $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(\ell)$
- **Hydrogen bromide with postassium hydroxide**
  Hydrogen bromide reacts with potassium hydroxide to form potassium bromide (the salt) and water. Potassium bromide is made up of K^+ cations from the base (KOH) and Br^- anions from the acid (HBr).
  $HBr(aq) + KOH(aq) \rightarrow KBr(aq) + H_2O(\ell)$
- **Hydrochloric acid with sodium hydrocarbonate**
  Hydrochloric acid reacts with sodium hydrocarbonate to form sodium chloride (the salt), water and carbon dioxide. Sodium chloride is made up of Na^+ cations from the base ($NaHCO_3$) and Cl^- anions from the acid (HCl).
  $HCl(aq) + NaHCO_3(aq) \rightarrow NaCl(aq) + H_2O(\ell) + CO_2(g)$

FACT
A **cation** is an ion (charged atom or molecule) with a positive (+) charge. An **anion** is an ion with a negative (-) charge.

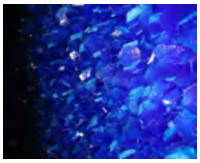
FACT

Salts are not just the table salt you put on your food. A salt is any compound made up of stoichiometrically equivalent amounts of cations and anions to make a neutral, ionic compound.

Magnesium sulfate heptahydrate ($MgSO_4.7H_2O$), commonly known as Epsom salt, can be used as a gel to treat aches and pains, as bath salts, and has many other uses.

FACT
Salts can come in many different colours.

Potassium permanganate ($KMnO_4$)



Copper sulfate ($CuSO_4$)

Nickel chloride ($NiCl_2$)

Sodium chromate (Na_2CrO_4)

Potassim dichromate ($K_2Cr_2O_7$)

You should notice that in the first two examples, the base contained OH^- ions, and therefore the products were a *salt* and *water*. NaCl (table salt) and KBr are both salts. In the third example, $NaHCO_3$ also acts as a base, despite not having OH^- ions. A salt is still formed as one of the products, but carbon dioxide (CO_2) is produced as well as water.

Informal experiment: Temperature changes in neutralisation reactions

Aim:
To investigate the temperature change associated with a neutralisation reaction.

Apparatus:

- 1 $mol.dm^{-3}$ solution of sodium hydroxide (NaOH), 1 $mol.dm^{-3}$ solution of hydrochloric acid (HCl)
- A thermometer, a beaker, a measuring cylinder

Method:
WARNING!

Concentrated acids and bases can cause serious burns. We suggest using gloves and safety glasses whenever you work with an acid or a base. Remember to add the acid to the water and to avoid sniffing any laboratory chemical. Handle all chemicals with care.

---

1. Pour 20 cm^3 of the sodium hydroxide solution into the beaker.
2. Measure the temperature.
3. Add 5 cm^3 of hydrochloric acid to the beaker using the measuring cylinder.
4. Repeat steps 2 and 3 quite quickly until the temperature no longer changes significantly.

Observations:

You should record your temperatures and the volume in a table (remember that the volume of the sodium hydroxide is constant at 20 cm^3):

Volume (HCl)	Temperature (°C)
0	room temperature

Discussion:
You should find that the reaction releases heat and so the temperature increases. After all the base has been neutralised the temperature should no longer increase. This is because the neutralisation reaction is *exothermic* (it releases heat). When all the base has been neutralised there is no reaction on the addition of more acid, and no more heat is released.

Neutralisation reactions are very important in every day life. Below are some examples:

- **Domestic uses**
  Calcium oxide (CaO) is used to neutralise acidic soil. Powdered limestone ($CaCO_3$) can also be used, but its action is much slower and less effective. These substances can also be used on a larger scale in farming and in rivers.
- **Biological uses**
  Hydrochloric acid (HCl) in the stomach plays an important role in helping to digest food. It is important to note that too much acid in the stomach may lead to the formation of ulcers in cases where the stomach lining is damaged (e.g. by an infection).

Antacids (which are bases) are taken to neutralise excess stomach acid, to prevent damage to the intestines. Examples of antacids are aluminium hydroxide, magnesium hydroxide ('milk of magnesia') and sodium bicarbonate ('bicarbonate of soda').

- **Industrial uses**

  Alkaline calcium hydroxide (limewater) is used to absorb harmful acidic SO_2 gas that is released from power stations and from the burning of fossil fuels.

FACT

Bee stings are acidic and have a pH between 5 and 5,5. They can be soothed by using substances such as calomine lotion, which is a mild alkali based on zinc oxide. Bicarbonate of soda, or soap, can also be used. The alkalis help to neutralise the acidic bee sting and relieve some of the itchiness.

WARNING!

Please do not use a base to neutralise an acid if you spill some on yourself during an experiment. A strong base can burn you as much as a strong acid. Rather wash the area thoroughly with water.

Activity: Acids and metal compounds

Research the reactions that occur when an **acid is added** to the following compounds:
(Your Grade 11 book will be helpful with this research)

- a metal
- a metal hydroxide
- a metal oxide
- a metal carbonate
- a metal hydrogen carbonate

Write a report which includes:

- The general equation (in words) of each reaction.
- A description of what is happening in this reaction.
- An example of this type of reaction in the form of a balanced equation.

Activity: The hazardous nature of acids and bases

Search for information about the following strong acids and bases:

- Hydrochloric acid (HCl)
- Sulfuric acid (H_2SO_4)
- Sodium hydroxide ($NaOH$)
- Potassium hydroxide (KOH)

Write a report which includes:

- The uses of these compounds in industry
- If applicable, the environmental waste that contains these compounds
- What the effect of a large spillage of these compounds would be

Worked example 8: Determining equations from starting materials

QUESTION

Magnesium carbonate ($MgCO_3$) is dissolved in nitric acid (HNO_3). Give the balanced chemical equation for this reaction.

SOLUTION

Step 1: What are the reactants?

An acid (HNO_3) and a metal carbonate ($MgCO_3$).

Step 2: What will the products be?

As this is the reaction of an acid and a metal carbonate the products will be a salt, water and carbon dioxide.

nitric acid + magnesium carbonate → salt + water + carbon dioxide

Step 3: What is the formula of the salt?

The cation will come from the metal carbonate (Mg^{2+}). The anion will come from the acid (NO_3^-). Due to the charges on the cation and anion there must be two NO_3^- for every one Mg^{2+}.

Therefore the formula for the salt will be: $Mg(NO_3)_2$.

Step 4: Write the equation for this reaction

$HNO_3(aq) + MgCO_3(s) \rightarrow Mg(NO_3)_2(aq) + H_2O(\ell) + CO_2(g)$

Step 5: Make sure that the equation is balanced

The equation is not balanced.

To balance this equation there needs to be two nitric acid molecules on the left hand side.

	Number on left	Number on right
H	1	2
N	1	2
O	6	9
Mg	1	1
C	1	1

$2HNO_3(aq) + MgCO_3(s) \rightarrow Mg(NO_3)_2(aq) + H_2O(\ell) + CO_2(g)$

The equation is now balanced.

	Number on left	Number on right
H	2	2
N	2	2
O	9	9
Mg	1	1
C	1	1

Worked example 9: Determining equations from starting materials

QUESTION

Hydroiodic acid (HI) is added to solid potassium hydroxide (KOH). Give the balanced chemical equation for this reaction.

SOLUTION

Step 1: What are the reactants?

An acid (HI) and a base (KOH).

Step 2: What will the products be?

As this is the reaction of an acid and base (which contains a hydroxide anion) the products will be a salt and water.

hydroiodic acid + potassium hydroxide → salt + water

Step 3: What is the formula of the salt?

The cation will come from the base (K^+). The anion will come from the acid (I^-). Due to the charges on the cation and anion there must be one K^+ for every one I^-. Therefore the formula for the salt will be: KI.

Step 4: Write the equation for this reaction

$HI(aq) + KOH(s) \rightarrow KI(aq) + H_2O(\ell)$

This can also be written:

$HI(aq) + KOH(s) \rightarrow K^+(aq) + I^-(aq) + H_2O(\ell)$

Step 5: Make sure that the equation is balanced

The equation is balanced.

Worked example 10: Determining equations from starting materials

QUESTION

Sulfuric acid (H_2SO_4) and ammonia (NH_3) are combined. Give the balanced chemical equation for this reaction.

SOLUTION

Step 1: What are the reactants?

An acid (H_2SO_4) and a base (NH_3).

Step 2: What will the products be?

As this is the reaction of an acid and a base (with no hydroxide anion), there will be a salt as a product. There may or may not be another product.

sulfuric acid + ammonia → salt (+ maybe another product)

Step 3: What is the formula of the salt?

The cation will come from the base (NH_4^+). The anion will come from the acid (SO_4^{2-}). Due to the charges on the cation and anion there must be two NH_4^+ for every one SO_4^{2-}. Therefore the formula for the salt will be: $(NH_4)_2SO_4$.

Step 4: Write the equation for this reaction so far

$H_2SO_4(aq) + NH_3(g) \rightarrow (NH_4)_2SO_4(aq)$ (+ maybe another product)

Step 5: Determine if there will be another product

There are no atom types that are not accounted for on both sides of the equation, therefore it is unlikely that there will be another product. If the equation can be balanced then there is no other product.

Step 6: Make sure that the equation is balanced

The equation is not balanced.

To balance this equation there needs to be two ammonia molecules on the left hand side.

	Number on left	Number on right
H	5	8
S	1	1
O	4	4
N	1	2

$H_2SO_4(aq) + 2NH_3(g) \rightarrow (NH_4)_2SO_4(aq)$

FACT
The term pH was first used by in 1909 by Søren Peter Lauritz Sørensen (a Danish biochemist). The p stood for *potenz* and the H for *hydrogen*. This translates to *power of hydrogen*.

FACT
Fizzy cooldrinks often have very low pH (are acidic).

Drink	pH
Coke	2,5
Diet coke	3,3
Pepsi	2,5
Diet pepsi	3,0
Sprite	3,2
7 Up	3,2
Diet 7 Up	3,7

The equation is now balanced.

	Number on left	Number on right
H	8	8
S	1	1
O	4	4
N	2	2

Exercise 9 – 4: Reactions of acids and bases

1. Write balanced equations for these acid and metal reactions:
   a) Hydrochloric acid and calcium
   b) Nitric acid and magnesium
2. Write balanced equations for these acid and metal hydroxide reactions:
   a) Hydrochloric acid and magnesium hydroxide
   b) Nitric acid and aluminium hydroxide
3. Write balanced equations for these acid and metal oxide reactions:
   a) Hydrochloric acid and aluminium oxide
   b) Sulfuric acid and magnesium oxide
4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27WT 2. 27WV 3. 27WW

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9.3 pH ESCPB

The pH scale ESCPC

The concentration of specific ions in solution determines whether the solution is acidic or basic. Acids and bases can be described as substances that either increase or decrease the concentration of hydrogen (H^+) or hydronium (H_3O^+) ions in a solution. An acid *increases* the hydrogen ion concentration in a solution, while a base *decreases* the hydrogen ion concentration.

pH is used to measure the concentration of H^+ ions ($[H^+]$) and therefore, whether a substance is acidic or basic (alkaline). Solutions with a pH of less than seven are acidic, while those with a pH greater than seven are basic (alkaline). The pH scale ranges from 0 to 14 and a pH of 7 is considered neutral.

DEFINITION: *pH*

pH is a measure of the acidity or alkalinity of a solution.

At the beginning of the chapter we mentioned that we encounter many examples of acids and bases in our day-to-day lives. The pH of solutions of some household acids and bases are given in Table 9.6.

Molecule	Found in	pH	Type
phosphoric acid	fizzy drinks	2,15	acid
tartaric acid	wine	2,95	acid
citric acid	lemon juice	3,14	acid
acetic acid	vinegar	4,76	acid
carbonic acid	fizzy drinks	6,37	acid
ammonia	cleaning products	11,5	base
ammonium hydroxide	cleaning products	11,63	base
sodium hydroxide	caustic soda	13	base

Table 9.6: The pH of solutions of acids and bases as found in common household items.

pH calculations ESCPD

pH can be calculated using the following equation: $pH = -\log[H^+]$

$[H^+]$ and $[H_3O^+]$ can be substituted for one another: $pH = -\log[H_3O^+]$

The brackets in the above equation are used to show **concentration** in $mol.dm^{-3}$.

Worked example 11: pH calculations

QUESTION

Calculate the pH of a solution where the concentration of hydrogen ions is 1×10^{-7} $mol.dm^{-3}$.

SOLUTION

Step 1: Determine the concentration of hydrogen ions

In this example, the concentration has been given: 1×10^{-7} $mol.dm^{-3}$

Step 2: Substitute this value into the pH equation and calculate the pH value

$pH = -\log[H^+] = -\log(1 \times 10^{-7}) = 7$

Worked example 12: pH calculations

QUESTION

In a 162 cm^3 solution of ethanoic acid, the following equilibrium is established:

$CH_3COOH(aq) + H_2O(\ell) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$

The number of moles of CH_3COO^- is found to be 0,001 mol. Calculate the pH of the solution.

SOLUTION

Step 1: Determine the number of moles of hydronium ions in the solution

According to the balanced equation for this reaction, the mole ratio of CH_3COO^- ions to H_3O^+ ions is 1:1, therefore the number of moles of these two ions in the solution will be the same.

So, $n(H_3O^+) = 0{,}001$ mol.

Step 2: Determine the concentration of hydronium ions in the solution

$$C\ (mol.dm^{-3}) = \frac{n\ (mol)}{V\ (dm^3)} \qquad V = 162\ cm^3 \times \frac{0{,}001\ dm^3}{1\ cm^3} = 0{,}162\ dm^3$$

TIP

The pH scale is a **log** scale. Remember from mathematics that a difference of one on a base 10 log scale (the one on your calculator) is equivalent to a multiplication by 10. That is:

$1 = \log(10)$

$2 = \log(100)$

$3 = \log(1000)$

$4 = \log(10000)$

So a change from a pH of 2 to a pH of 6 represents a very large change in the H^+ concentration.

TIP

Important: It may be useful to know that for calculations involving the pH scale, the following equations can also be used:

$[H^+][OH^-] = 1 \times 10^{-14}$

$[H_3O^+][OH^-] = 1 \times 10^{-14}$

$pH = 14 - p[OH^-]$

$pH = 14 - (-\log[OH^-])$

FACT

A build up of acid in the human body can be very dangerous. **Lactic acidosis** is a condition caused by the buildup of lactic acid in the body. It leads to acidification of the blood (acidosis) and can make a person very ill. Some of the symptoms of lactic acidosis are deep and rapid breathing, vomiting, and abdominal pain. In the fight against HIV, lactic acidosis is a problem. One of the antiretrovirals (ARV's) that is used in anti-HIV treatment is Stavudine (also known as Zerit or d4T). One of the side effects of Stavudine is lactic acidosis, particularly in overweight women. If it is not treated quickly, it can result in death.

FACT

Litmus paper can be used as a pH indicator. It is sold in strips. Purple litmus paper will become red in acidic conditions and blue in basic conditions. Blue litmus paper is used to detect acidic conditions, while red litmus paper is used to detect basic conditions.

$$[H_3O^+] = \frac{0{,}001 \text{ mol}}{0{,}162 \text{ dm}^3} = 0{,}0062 \text{ mol.dm}^{-3}$$

Step 3: Substitute this value into the pH equation and calculate the pH value

$\text{pH} = -\log[H^+] = -\log[H_3O^+] = -\log(0{,}0062) = 2.21$

Understanding pH is very important. In living organisms, it is necessary to maintain a constant pH in the optimal range for that organism, so that chemical reactions can occur.

pH	1	6	7	8	13
$[H^+]$	1×10^{-1}	1×10^{-6}	1×10^{-7}	1×10^{-8}	1×10^{-13}
$[OH^-]$	1×10^{-13}	1×10^{-8}	1×10^{-7}	1×10^{-6}	1×10^{-1}
Solution	**strongly acidic**	weakly acidic	neutral	weakly basic	**strongly basic**

acidic solution **basic solution**

Table 9.7: The concentration of $[H^+]$ and $[OH^-]$ ions in solutions with different pH.

In agriculture, it is important for farmers to know the pH of their soils so that they are able to plant the right kinds of crops. The pH of soils can vary depending on a number of factors, such as rainwater, the kinds of rocks and materials from which the soil was formed and also human influences such as pollution and fertilisers. The pH of rain water can also vary, and this too has an effect on agriculture, buildings, water courses, animals and plants. Rainwater is naturally acidic because carbon dioxide in the atmosphere combines with water to form carbonic acid. Unpolluted rainwater has a pH of approximately 5,6. However, human activities can alter the acidity of rain and this can cause serious problems such as acid rain.

Exercise 9 – 5: Calculating pH

1. Calculate the pH of each of the following solutions:
   (Tip: $[H_3O^+][OH^-] = 1 \times 10^{-14}$ can be used to determine $[H_3O^+]$)
   a) A KOH solution with a 0,2 mol.dm^{-3} concentration of OH^-.
   b) An aqueous solution with a $1{,}83 \times 10^{-7}$ mol.dm^{-3} concentration of HCl molecules at equilibrium ($K_a = 1{,}3 \times 10^6$)
2. What is the concentration of H_3O^+ ions in a solution with a pH of 12?
3. In a typical sample of seawater the concentration the hydronium (H_3O^+) ions is 1×10^{-8} mol.dm^{-3}, while the concentration of the hydroxide (OH^-) ions is 1×10^{-6} mol.dm^{-3}.
   a) Is the seawater acidic or basic?
   b) What is the pH of the seawater?
   c) Give a possible explanation for the pH of the seawater.
4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27WX 2. 27WY 3. 27WZ

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K_w ESCPF

TIP
A neutralisation reaction **does not imply** that the pH is neutral (7).

Water ionises to a very small extent: $H_2O(\ell) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$

This type of reaction (the transfer of a proton between identical molecules) is known as **auto-protolysis**. This reaction is also known as the **auto-ionisation** of water and the ions formed are a conjugate acid and base pair of water:

conjugate pair

$$\underset{\text{acid}}{H_2O(\ell)} + \underset{\text{base}}{H_2O(\ell)} \rightleftharpoons \underset{\text{conjugate acid}}{H_3O^+(aq)} + \underset{\text{conjugate base}}{OH^-(aq)}$$

conjugate pair

DEFINITION: *Auto-protolysis and auto-ionisation of water*

Auto-protolysis is the transfer of a proton between two of the same molecules. The auto-ionisation of water is one example of auto-protolysis.

K_w is the equilibrium constant for this process: $K_w = [H_3O^+][OH^-]$

At 25 °C: $[H_3O^+] = [H^+] = 1 \times 10^{-7}$, therefore: $K_w = 1 \times 10^{-14}$ at 25 °C

Salt hydrolysis ESCPG

Does neutralisation mean that the pH of the solution is 7? No. At the equivalence point of a reaction, the pH of the solution need not be 7. This is because of the interaction of the salt (formed by the reaction) and water.

At the equivalence point of an acid-base neutralisation reaction there is salt and water. The ions of water interact with the salt present and form a small quantity of excess hydronium ions (H_3O^+) or hydroxide ions (OH^-). This leads to pH values that are not equal to 7.

A simple rule for determining the likely pH of a solution is as follows:

- A **strong** acid + **strong** base form a **neutral** salt and water **solution**:
  $\rightarrow$ pH = 7.
  $H_2SO_4(\ell) + NaOH(s) \rightarrow \mathbf{Na_2SO_4(aq)} + H_2O(\ell)$
- A **weak** acid + **strong** base form a **weak basic** salt and water **solution**:
  $\rightarrow$ pH = approximately 9.
  $HF(\ell) + NaOH(s) \rightarrow \mathbf{NaF(aq)} + H_2O(\ell)$
- A **strong** acid + **weak** base form a **weak acidic** salt and water **solution**:
  $\rightarrow$ pH = approximately 5.
  $H_2SO_4(\ell) + NH_3(\ell) \rightarrow \mathbf{(NH_4)_2SO_4(aq)} + H_2O(\ell)$

acid	base	solution	approximate pH
strong	strong	**neutral**	7
weak	strong	**weak basic**	9
strong	weak	**weak acidic**	5

Table 9.8: The approximate pH of neutralisation reaction solutions based on the strength of the acid and base used.

FACT
The universal indicator changes colour from **red** in **strongly acidic** solutions through to **purple** in **strongly basic** solutions.

Indicators ESCPH

A titration is a process for determining, with precision, the concentration of a solution with unknown concentration. The theory behind titrations will be discussed later in this chapter. An indicator is used to show the scientist carrying out the reaction exactly when the reaction has reached completion.

Experiment: Indicators

Aim:

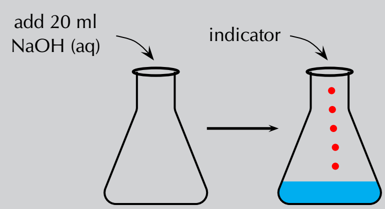
To investigate the use of an indicator in an acid-base reaction.

Apparatus and materials:

- one volumetric flask, one conical flask, one pipette, a piece of white paper or a white tile
- A 1 $mol.dm^{-3}$ solution of sodium hydroxide ($NaOH$), a 1 $mol.dm^{-3}$ solution of hydrochloric (HCl), an indicator

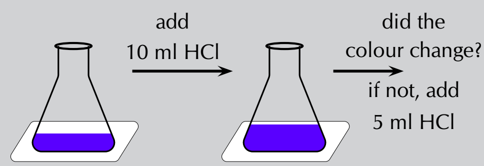
Method:

1. Measure 20 ml of the sodium hydroxide solution into a conical flask. Add a few drops of the indicator.



2. In this experiment colour change is very important. So place the conical flask on a piece of white paper or a white tile to make any colour change easier to observe.
3. Slowly add 10 ml of hydrochloric acid. If there is a colour change, then stop. If there is no colour change add another 5 ml.

   Continue adding 5 ml increments until you notice a colour change.



Observations and discussion:

The solution changes colour after a certain amount of hydrochloric acid is added. This is because the solution now contains more acid than base and has therefore become acidic. It can be concluded that the indicator is one colour in a basic solution and a different colour in an acidic solution.

Indicators are chemical compounds that change colour depending on whether they are in an acidic or a basic solution. A titration requires an indicator that will respond to the change in pH with a sensitive and quick colour change. Typical indicators used in titrations are given in Table 9.9.

Titration type	Preferred indicator	Colour of acid	Colour of end-point	Colour of base	pH range
strong acid + strong base	bromothymol blue	yellow	green	blue	6,0 - 7,6
weak acid + strong base	phenolphthalein	colourless	faint pink	pink	8,3 - 10,0
strong acid + weak base	bromocresol green	yellow	green	blue	3,8 - 5,4

Table 9.9: Some typical indicators for typcial titrations.

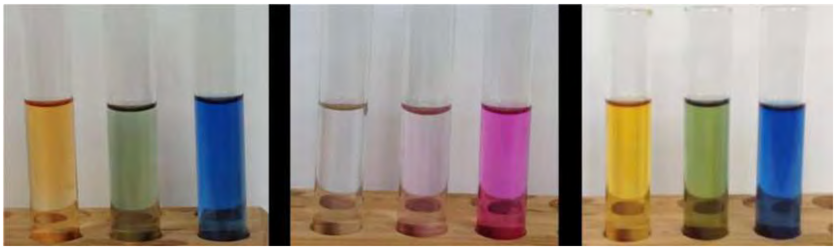


Figure 9.3: Some typical indicators for typcial titrations: bromothymol blue (left), phenolphthalein (centre), and bromocresol green (right).

Indicators change colour (Figure 9.3) according to where the H is:

$$\text{HBromothymol blue} \rightleftharpoons \text{H}^+(\text{aq}) + \text{Bromothymol blue}^-(\text{aq})$$

yellow blue

So, when an acid is added to aqueous bromothymol blue there will be extra H^+ ions. The equilibrium will shift (remember le Chatalier's principle) to decrease the number of H^+ ions. That is, to the left. If sufficient acid is added, the entire solution will become acidic. This means there will be more *HBromothymol blue* than *bromothymol blue*$^-$ and the solution will become yellow.

TIP
Revise Grade 11 Acids and Bases for more information on plants and foods that can be used as indicators.

TIP
Notice that the pH range for colour change of the indicator used should match the approximate pH expected for that type of titration (see Table 9.8):

solution	pH	pH range
neutral	7	6,0 - 7,6
weak basic	9	8,3 - 10,0
weak acidic	5	3,8 - 5,4

9.4 Titrations ESCPJ

What are titrations? ESCPK

The neutralisation reaction between an acid and a base can be very useful. If an acidic solution of known concentration (a standard solution) is added to a basic (alkaline) solution of unknown concentration until the solution is exactly neutralised (i.e. there is only salt and water), it is possible to calculate the exact concentration of the unknown solution. It is possible to do this because, at the exact point where the solution is neutralised, stoichiometrically equivalent mole amounts of acid and base have reacted with each other.

DEFINITION: *Titration*

The method used to determine the concentration of a known substance using another, standard, solution.

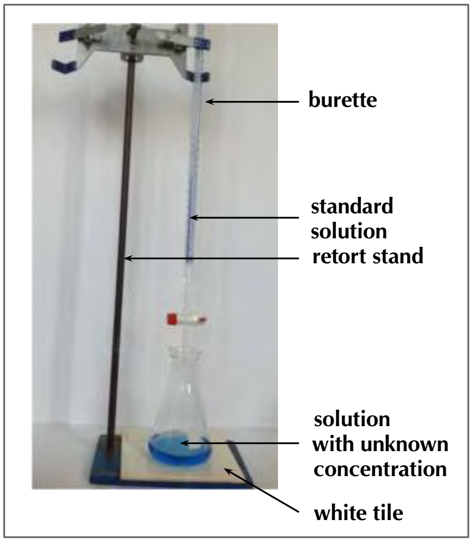
In a **titration**: a known volume of a standard solution (**A**) is added to a known volume of a solution with unknown concentration (**B**). The concentration of **B** can then be determined.

Acids and bases are commonly used in titrations, and the point of neutralisation is called the **end-point** of the reaction. If you have an indicator that changes colour in the range of the end-point pH then you will be able to see when the end-point has occurred. Another name for a titration is **volumetric analysis**.

Titration calculations ESCPM

So how exactly can a titration be carried out to determine an unknown concentration? Look at the following steps to help you to understand the process.

1. A carefully measured volume of the solution with unknown concentration is put into a conical flask.
2. A few drops of a suitable indicator is added to this solution (bromothymol blue and phenolphthalein are common indicators, refer to Table 9.9).
3. The conical flask is placed on a white tile or piece of paper (to make colour changes easier to see).
4. A volume of the standard solution (known concentration) is put into a burette (a measuring device) and is slowly added to the solution in the flask, drop by drop.



5. At some point, adding one more drop will change the colour of the unknown solution to the colour of the end-point of the reaction. Remember the colour changes from Figure 9.3.
6. Record the volume of standard solution that has been added up to this point.
7. Use the information you have gathered to calculate the exact concentration of the unknown solution. Worked examples are given to walk you through this step.
8. Note that adding more solution once the end-point has been reached will result in a colour change from the end-point colour to that of the acid (if the solution in the conical flask is a base) or of the base (if the solution in the conical flask is an acid).

When you are busy with these calculations, you will need to remember the following:

1 dm^3 = 1 litre = 1000 ml = 1000 cm^3, therefore dividing cm^3 by 1000 will give you an answer in dm^3.

Some other terms and equations which will be useful to remember are shown below:

- concentration of a solution is measured in $mol.dm^{-3}$
- moles (mol) = concentration ($mol.dm^{-3}$) $\times$ volume (dm^3)
- concentration $= \frac{\text{moles}}{\text{volume}}$ $\qquad C\ (mol.dm^{-3}) = \frac{n\ (mol)}{V\ (dm^3)}$
- remember to make sure all the units are correct in your calculations

Worked example 13: Titration calculations

QUESTION

Given the equation: $NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(\ell)$

25 cm^3 of sodium hydroxide solution was pipetted into a conical flask and titrated with 0,2 $mol.dm^{-3}$ hydrochloric acid. Using a suitable indicator, it was found that 15 cm^3 of acid was needed to neutralise the base. Calculate the concentration of the sodium hydroxide.

SOLUTION

Step 1: Make sure that the equation is balanced

There are equal numbers of each type of atom on each side of the equation, so the equation is balanced.

Step 2: Write down all the information you know about the reaction, converting to the correct units

NaOH: $V = 25\ cm^3 \times \frac{0{,}001\ dm^3}{1\ cm^3} = 0{,}025\ dm^3$

HCl: $V = 15\ cm^3 \times \frac{0{,}001\ dm^3}{1\ cm^3} = 0{,}015\ dm^3$ $\qquad C = 0{,}2\ mol.dm^{-3}$

Step 3: Calculate the number of moles of HCl that are added

$C = \frac{n}{V}$ $\qquad$ Therefore, $n(HCl) = C \times V$

$n(HCl) = 0{,}2\ mol.dm^{-3} \times 0{,}015\ dm^3 = 0{,}003\ mol$

0,003 moles of HCl are required to neutralise the base.

Step 4: Calculate the number of moles of sodium hydroxide in the reaction

Look at the equation for the reaction: the molar ratio of HCl:NaOH is 1:1.

So for every mole of HCl, there is one mole of NaOH that is involved in the reaction. Therefore, if 0,003 mol HCl is required to neutralise the solution, 0,003 mol NaOH must have been present in the sample of the unknown solution.

Step 5: Calculate the concentration of the sodium hydroxide solution

$C(NaOH) = \frac{n}{V} = \frac{0{,}003\ mol}{0{,}025\ dm^3} = 0{,}12\ mol.dm^{-3}$

The concentration of the NaOH solution is 0,12 $mol.dm^{-3}$

Worked example 14: Titration calculations

QUESTION

10 g of solid sodium hydroxide is dissolved in 500 cm^3 water. Using titration, it was found that 20 cm^3 of this solution was able to completely neutralise 10 cm^3 of a sulfuric acid solution. Calculate the concentration of the sulfuric acid.

SOLUTION

Step 1: Write a balanced equation for the titration reaction

The reactants are sodium hydroxide (NaOH) and sulfuric acid (H_2SO_4). The base has a hydroxide anion (OH^-), therefore the products will be a salt and water.

The cation for the salt (Na^+) will come from the base. The anion for the salt (SO_4^{2-}) will come from the acid. There must be 2 Na^+ cations for every one SO_4^{2-} and the salt will be Na_2SO_4.

$H_2SO_4(aq) + NaOH(aq) \rightarrow Na_2SO_4(aq) + H_2O(\ell)$

To balance the equation we need to multipy the number of sodium hydroxide molecules and the water molecules by two.

$H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(\ell)$

Step 2: Calculate the concentration of the sodium hydroxide solution

The total volume that the 10 g was dissolved in must be used to calculate the concentration.

$$V = 500\ cm^3 \times \frac{0,001\ dm^3}{1\ cm^3} = 0,5\ dm^3$$

$$M(NaOH) = 23,0 + 16,0 + 1,01 = 40,01\ g.mol^{-1}$$

$$n(NaOH) = \frac{m}{M} = \frac{10\ g}{40,01\ g.mol^{-1}} = 0,25\ mol$$

$$C(NaOH) = \frac{n}{V} = \frac{0,25\ mol}{0,5\ dm^3} = 0,50\ mol.dm^{-3}$$

Step 3: Calculate the number of moles of sodium hydroxide that were used in the neutralisation reaction

Remember that only 20 cm^3 of the sodium hydroxide solution is used:

$$V = 20\ cm^3 \times \frac{0,001\ dm^3}{1\ cm^3} = 0,02\ dm^3 \qquad C = \frac{n}{V}, \text{ therefore } n = C \times V$$

$$n = 0,50\ mol.dm^{-3} \times 0,02\ dm^3 = 0,01\ mol$$

Step 4: Calculate the number of moles of sulfuric acid that were neutralised

According to the balanced chemical equation, the mole ratio of NaOH to H_2SO_4 is 2:1. There are 2 moles of NaOH for every 1 mole of H_2SO_4.

$$n(H_2SO_4) = \frac{0,01\ mol}{2} = 0,005\ mol.$$

Step 5: Calculate the concentration of the sulfuric acid solution

Remember that 10 cm^3 of the sulfuric acid solution is neutralised.

$$V(H_2SO_4) = 10\ cm^3 \times \frac{0,001\ dm^3}{1\ cm^3} = 0,01\ dm^3$$

$$C = \frac{n}{V} = \frac{0,005\ mol}{0,01\ dm^3} = 0,5\ mol.dm^{-3}$$

Formal experiment: Preparation of a standard solution

Aim:

To prepare a standard solution of sodium hydroxide.

Apparatus and materials:

- one 250 cm^3 volumetric flask, a beaker, a funnel, a plastic dropper
- A weighing scale, a spatula, a clean piece of paper
- solid sodium hydroxide

Method:

WARNING!

Concentrated bases can cause serious burns. We suggest using gloves and safety glasses whenever you work with a base. Handle all chemicals with care.

1. Place the paper on the scale and tare (zero) the scale.
2. Weigh 5,00 g of solid sodium hydroxide onto the paper.
3. Transfer the sodium hydroxide to the beaker and dissolve it in a minimal amount of water.
4. Place the funnel in the mouth of the volumetric flask and use it to transfer the solution from the beaker to the flask, be careful not to spill.
5. Pour another 5 cm^3 of water into the beaker, swirl gently, and pour this into the flask through the funnel as well.
6. Repeat step 5 another two times.
7. Pour 5 cm^3 of water through the funnel into the flask.
8. Use the plastic dropper to pour water down the inside of the volumetric flask.
9. Fill the volumetric flask to the mark with water. Use the plastic dropper to fill the last few centimeters. Remember to lower yourself so that you are looking directly at the mark when adding the final drops.
10. Cap the flask and shake well. You now have a standard solution.

Questions:

- What is the concentration of this standard solution?
- What is the purpose of rinsing the beaker and funnel with water?
- What is the purpose of rinsing the inside of the volumetric flask with water?
- Why do you need to make sure your eyes are level with the mark when adding the final few drops of water?

Experiment: Titrations

Aim:
To determine the concentration of acetic acid (ethanoic acid) in a sample of vinegar.

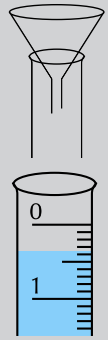
Apparatus and materials:

- 4 conical flasks, one pipette, one burette, one small funnel, one beaker
- a retort stand, a white piece of paper or a white tile
- the standard NaOH solution prepared in the previous experiment, white vinegar, phenolphthalein

Method:

WARNING!

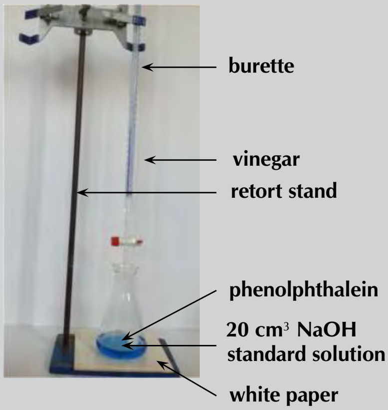
Concentrated acids and bases can cause serious burns. We suggest using gloves and safety glasses whenever you work with an acid or a base. Handle all chemicals with care.



1. Clamp the burette to the retort stand and place the small funnel on top.
2. Making sure the burette is closed, carefully add the vinegar to the burette (lift the funnel slightly while doing this) until the burette is full.
3. Place a beaker below the burette and carefully let some of the vinegar run into it. This will ensure that there are no air bubbles in the burette. The top of the vinegar should now be between the 0 and 1 cm^3 marks. Record the value to the second decimal place.
4. Use your pipette to measure 20 cm^3 of the sodium hydroxide standard solution into a conical flask.
5. Add 3 - 4 drops of phenolphthalein indicator to the conical flask. What colour is the solution?
6. Do a rough titration experiment by adding the vinegar to the conical flask quickly, and constantly swirling the conical flask. Stop as soon as the colour of the solution changes, and remains changed after swirling. Make a note of the reading on the burette at this point and determine the volume added:

   (V(final) - V(initial)).

   If the colour change does not remain when the flask is swirled, add more vinegar until the colour remains.



7. Repeat steps 2 - 5 with a second conical flask.
8. Quickly add vinegar to the conical flask until you have added 2 cm^3 less than your rough titration volume. There should be no lasting colour change at this point.
9. Drop by drop (slowly and carefully), add vinegar. Swirl between each drop and, if necessary, rinse the sides of the flask with water. When the solution changes colour and remains that new colour, make a note of the volume on the burette. Remember that the volume titrated is: (final volume reading) - (the initial volume reading).
10. For accuracy you should repeat steps 7 - 9 until you have three readings with a difference of no more than 0,1 cm^3.

Observations and questions:

- What colour was the sodium hydroxide solution when the phenolphthalein was added?
- What was the colour when enough acid was added?

Fill in the details of this experiment on a table:

Titration	Initial volume (cm^3)	Final volume (cm^3)	Volume (cm^3)
Rough			
1			
2			
3			

- From the table determine the average titration volume for this experiment and use that value in the rest of your calculations.
- Using the previous worked examples, determine the concentration of acetic acid in the sample of vinegar. Remember that you have the following information:
  - the volume of vinegar
  - the volume of the sodium hydroxide solution
  - the concentration of the sodium hydroxide solution
- The balanced chemical equation for this reaction is:
  $CH_3COOH(aq) + NaOH(aq) \rightarrow CH_3COONa(aq) + H_2O(\ell)$

Experiment: Titrations

Aim:
To determine the concentration of a sodium hydroxide solution of unknown concentration.

Apparatus and materials:

- one 250 cm^3 volumetric flask, 4 conical flasks, one pipette, one burette, one small funnel, two beakers, one plastic dropper
- a retort stand, a weighing scale, a spatula, a clean piece of paper
- solid sodium hydroxide (NaOH), solid oxalic acid ($H_2C_2O_4$), phenolphthalein

Method:
WARNING!

Concentrated acids and bases can cause serious burns. We suggest using gloves and safety glasses whenever you work with an acid or a base. Handle all chemicals with care.

1. Prepare a standard solution using 11,00 g of oxalic acid in the 250 cm^3 volumetric flask.
2. Label one of the beakers *NaOH*. Use your pipette to measure 100 cm^3 of water into the beaker. Add approximately 4 g of NaOH to the beaker and stir.
3. Clamp the burette to the retort stand and place the small funnel on top.
4. Making sure the burette is closed, carefully add the oxalic acid standard solution to the burette (lift the funnel slightly while doing this) until the burette is full.
5. Place the clean beaker below the burette and carefully let some of the oxalic acid solution run into it. This will ensure that there are no air bubbles in the burette. The top of the solution should now be between the 0 and 1 cm^3 marks. Record the value to the second decimal place.
6. Use your pipette to measure 20 cm^3 of the NaOH solution into a conical flask.
7. Add 3 - 4 drops of phenolphthalein indicator to the conical flask. What colour is the solution?

8. Do a rough titration experiment by adding the oxalic acid to the conical flask quickly, and constantly swirling the conical flask. Stop as soon as the colour of the solution changes, and remains changed after swirling. Make a note of the reading on the burette at this point and determine the volume added:

   V(final) - V(initial).

   If the colour change does not remain when the flask is swirled, add more oxalic acid until the colour remains.
9. Repeat steps 4 - 7 with a second conical flask.
10. Quickly add oxalic acid to the conical flask until you have added 2 cm^3 less than your rough titration volume. There should be no lasting colour change at this point.
11. Drop by drop (slowly and carefully), add oxalic acid. Swirl between each drop and, if necessary, rinse the sides of the flask with water. When the solution changes colour, and remains that new colour, make a note of the volume on the burette.

    Remember that the volume titrated is: (final volume reading) - (the initial volume reading).
12. For accuracy, you should repeat steps 9 - 11 until you have three readings with a difference of no more than 0,1 cm^3.

Observations and questions:

- What colour was the sodium hydroxide solution when the phenolphthalein was added?
- What was the colour when enough acid was added?

Fill in the details of this experiment on a table like the one below:

Titration	Initial volume (cm^3)	Final volume (cm^3)	Volume (cm^3)
Rough			
1			
2			
3			

- From the table, determine the average titration volume for this experiment and use that value in the rest of your calculations.
- Using the previous worked examples, determine the concentration of the sodium hydroxide solution. Remember that you have the following information:
  - the volume of sodium hydroxide solution
  - the volume of oxalic acid solution
  - the concentration of the oxalic acid solution.
- The balanced chemical equation for this reaction is:

  $H_2C_2O_4(aq) + 2NaOH(aq) \rightarrow Na_2C_2O_4(aq) + 2H_2O(\ell)$

Exercise 9 – 6: Acids and bases

1. A learner is asked to prepare a *standard solution* of the weak acid, oxalic acid ($H_2C_2O_4$), for use in a titration. The volume of the solution must be 500 cm^3 and the concentration must be 0,2 $mol.dm^{-3}$.

   a) Calculate the mass of oxalic acid which the learner must dissolve to make up the required standard solution.

b) The learner titrates this 0,2 $mol.dm^{-3}$ oxalic acid solution against a solution of sodium hydroxide. He finds that 40 cm^3 of the oxalic acid solution completely neutralises 35 cm^3 of the sodium hydroxide solution. Calculate the concentration of the sodium hydroxide solution.

2. 25,0 cm^3 of a 0,1 $mol.dm^{-3}$ standard solution of sodium carbonate was used to neutralise 35,0 cm^3 of a solution of hydrochloric acid.

   a) Write a balanced chemical equation for the reaction.

   b) Calculate the concentration of the acid. (DoE Grade 11 Exemplar, 2007)

3. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27X2 2. 27X3

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9.5 Applications of acids and bases ESCPN

The production of chlorine ESCPP

The chlorine-alkali (chloralkali) industry is an important part of the chemical industry for the production of **chlorine** and **sodium hydroxide**. The most common method involves the electrolysis of a concentrated aqueous solution of sodium chloride (NaCl), which is known as **brine**. For more information on electrolysis see Chapter 13.

The chemical reactions that take place in this process are:

$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$ $2H_2O(\ell) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$

There are also Na^+ ions from the NaCl in the solution. If the products are kept separate (to prevent a reaction between the chlorine and hydroxide) the Na^+ will react with the hydroxide ions making the overall reaction as follows:

$2NaCl(aq) + 2H_2O(\ell) \rightarrow Cl_2(g) + H_2(g) + 2NaOH(aq)$

The uses of chlorine include:

- the purification of water
- as a disinfectant
- in the production of:
  - hypochlorous acid (used to kill bacteria in drinking water), chloroform, carbon tetrachloride
  - paper, food
  - antiseptics, insecticides, medicines, textiles
  - paints, petroleum products, solvents, plastics (such as polyvinyl chloride)

The uses of sodium hydroxide include:

- making soap and other cleaning agents
- purification of bauxite (the ore of aluminium)
- making paper
- making rayon (artificial silk)

If the chlorine and hydroxide ions are not kept separate (i.e. are allowed to react), and the temperature is kept below 60 °C, then the following occurs:

$Cl_2(g) + 2OH^-(aq) \rightarrow Cl^-(aq) + ClO^-(aq) + H_2O(\ell)$

The balanced chemical equation with the inclusion of the Na^+ ions is:

$Cl_2(g) + 2NaOH(aq) \rightarrow NaCl(aq) + NaClO(aq) + H_2O(\ell)$

If the temperature is above 60 °C and mixing occurs, then:

$3Cl_2(g) + 6OH^-(aq) \rightarrow 5Cl^-(aq) + ClO_3(aq) + 3H_2O(\ell)$

And the balanced chemical equation including Na^+ ions is:

$3Cl_2(g) + 6NaOH(aq) \rightarrow 5NaCl(aq) + NaClO_3(aq) + 3H_2O(\ell)$

The uses of NaClO (sodium hypochlorite) include:

- use in bleaches, disinfectants, and water treatments
- use during root canal surgery and to neutralise nerve agents

The uses of $NaClO_3$ (sodium chlorate) include:

- making chlorine dioxide
- use as a herbicide
- generating oxygen in chemical oxygen generators

If calcium chloride was used instead of sodium chloride, then the products would be calcium chloride, calcium hypochlorite and calcium chlorate. Similarly, if potassium chloride was used, the products would be potassium chloride, potassium hypochlorite and potassium chlorate.

The chemistry of hair and hair products ESCPQ

Hair is primarily made up of a protein called **keratin**. Keratin is a macromolecule and consists largely of the amino acid **cystine** (an organic acid, see Figure 9.4).

Amino acids are composed of an amine group (Figure 9.5 (a)) and a carboxylic acid group (Figure 9.5 (b)) connected by a carbon chain.

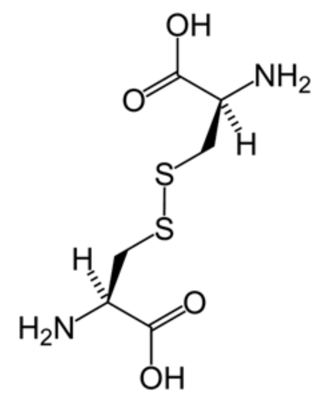


Figure 9.4: The main amino acid found in keratin: **cystine**.

Proteins give hair a natural acidic pH of between 4 and 5. The following hair products alter this pH and do *controlled damage* to get the desired effect.

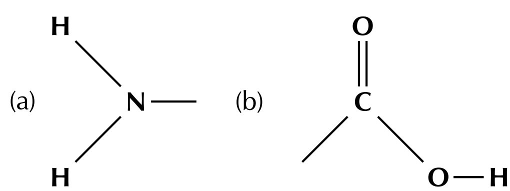


Figure 9.5: **a)** An amine functional group and **b)** a carboxylic acid functional group. Both of these are found in amino acids.

Permanent waving application

Another name for a permanent waving application is a **perm**. Ammonium thioglycolate is also known as a *perm salt*. The reversible reaction of ammonium thioglycolate (reactant) to form thioglycolic acid and free ammonia (product) is given below:

$$HSCH_2COO^-(aq) + NH_4^+(aq) \rightleftharpoons HSCH_2COOH(aq) + NH_3(aq)$$

There are four common steps in the perming process:

1. The basic ammonia swells the hair and makes it permeable.
2. The thioglycolic acid reduces the sulfur-sulfur bonds in the **cystine** molecules in your hair. This reduces cystine to **cysteine** (Figure 9.6) and allows the hair structure to be changed.

   This reaction occurs faster at higher temperatures, and the hair is generally heated at this time (e.g., on a hot curling iron).

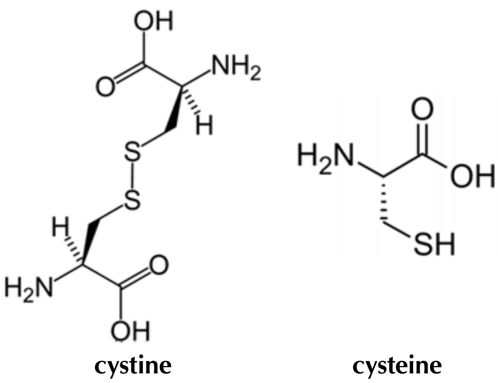
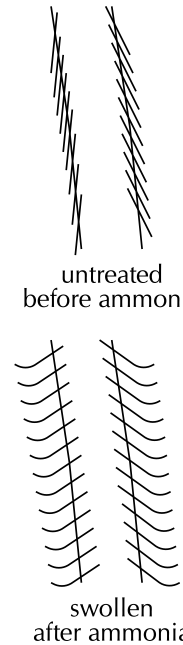


Figure 9.6: **Cystine** (left) is reduced to form **cysteine** (right).

3. The ammonium thioglycolate solution is washed out of the hair after the desired bond breaking has occurred.
4. The hair is treated with hydrogen peroxide (H_2O_2), which acts as a neutraliser. It oxidises the cysteine back to cystine (re-forming the sulfur-sulfur bonds). These bonds will form in the new positions formed during the ammonium thioglycolate application. These bonds then fix the hair in this new structure.

TIP

A strand of hair



FACT

It is important to use the neutraliser as well as wash the hair thoroughly after a perm application. Excess exposure to the ammonium thioglycolate can cause scalp irritation, as well as permanent hair damage.

Hair relaxers

Hair relaxers work by doing *controlled damage* to your hair in much the same way as perms do. They reduce the cystine to cysteine and allow the hair structure to be changed (Figure 9.6).

There are two main types of hair relaxers: **lye** and **no-lye**. The **lye** hair relaxers contain sodium hydroxide (NaOH) and have a very high pH (in the range 12 - 14). These types of hair relaxers break bonds and strip the hair of its natural oils. Often heating is also used and this damages the hair even more.

Due to increasing worry about the damage caused by lye hair relaxers, companies have produced **no-lye** hair relaxers. These still contain basic compounds and are still caustic, but their pH is not as high (pH in the range 9 - 11).

These products either use calcium hydroxide and guanidine carbonate (to form guanidine hydroxide, Figure 9.7) or lithium hydroxide. These are bases and act in a similar manner to sodium hydroxide. Although they are milder on hair than sodium hydroxide, they are still damaging. Another type of no-lye hair relaxer contains ammonium thioglycolate (the perm salt).

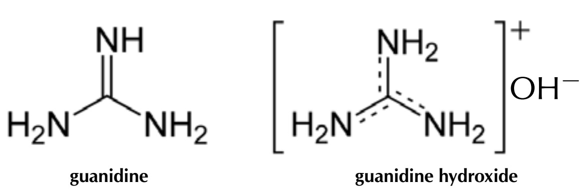


Figure 9.7: Guanidine (left) and guanidine hydroxide (right). Guanidine hydroxide is formed from guanidine carbonate and calcium hydroxide.

FACT
NaOH is used in drain cleaning products to dissolve hair, as well as in procedures that remove body hair.

FACT
Remember that the pH scale goes from 0 - 14.

Some comparisons between lye and no-lye hair relaxers are given in Table 9.10.

	Lye	No-lye
pH	12 - 14	9 - 11
Benefits	quick application time	more gentle on scalp
Problems	high pH, very damaging to scalp and hair	often left on too long, dries hair due to calcium build up

Table 9.10: Lye versus no-lye hair relaxers.

Colouring hair

There are many different types of hair colourants, ranging from permanent hair dye to temporary hair dye. Each of them works in a slightly different way.

- **Permanent**

  In permanent hair dyes there are three main ingredients: a diamino compound (that means two of the amine groups shown in Figure 9.5 (a)); a coupling agent; and an oxidising agent.

  The oxidising agent is generally hydrogen peroxide (H_2O_2). It oxidises the diamino compound to a state where it can bond with the coupling agent. This process also lightens the hair so that the dye will work more effectively. This new compound is then oxidised again to form the final dye.

  Permanent dyes are basic with a pH between 9 and11 (usually ammonia, NH_3). The purpose of the base is to open up the hair and allow the small dye molecules to penetrate.

- **Demi-permanent**

  These dyes use ethanolamine or sodium carbonate as a base. These are less harsh bases than ammonia (pH between 8 and 9), and so do not allow the dye to penetrate as far into the hair.

  They still contain a hydrogen peroxide developer that lightens the hair slightly. However, they cannot colour dark hair to a lighter colour. The hair colour will look more natural though, as the natural colour of the hair will affect the final dye. Demi-permanent dyes last a long time (at least 12 washes), but will eventually wash out.

  Demi-permanent hair dyes do not damage the hair as much as permanent dyes.

- **Semi-permanent**

  These dyes contain molecules that are small enough to partially penetrate the hair without the need of a base to open the hair. They last for about 4 - 5 washes and have, at most, very low levels of hydrogen peroxide or ammonia. They will not lighten hair and the final colour depends a lot on the original colour and the porosity of the hair.

- **Temporary**

  These are generally bright dyes. They are composed of large dye molecules, which are unable to penetrate the hair. These large molecules *stick* to the hair and usually can be removed with just one wash. If the hair is already dry or damaged the colour may last longer, as the damaged hair is more open and allows the dye molecules to penetrate.

It is worth noting that some of the possible side-effects of using perming applications, hair relaxers and hair colourants are:

- scalp irritation, skin burns, and permanent scarring
- deep ulcerations, dermatitis, skin drying and cracking
- irreversible baldness, eye damage or blindness
- weak, broken, or damaged hair

Experiment: The effect of drain cleaner on hair

Aim:

To determine the effect of caustic soda or drain cleaner on hair.

Apparatus:

- A pile of hair from your hairbrush
- A drain cleaning product or caustic soda flakes
- A watch glass

WARNING!

Be careful when handling any drain cleaning products. They are very basic and can burn your skin. We suggest using gloves and safety glasses as well as protective clothing. Handle all chemicals with care.

Method:

1. Collect the hair from your hair brush for a week.
2. Place the hair on a watch glass.
3. If you have a liquid drain cleaning product perform step **a**. If you have caustic soda perform step **b**.
   a) Carefully pour a drain cleaning product onto the pile of hair and observe.
   b) Take the solid caustic soda and put a few flakes onto the hair. Boil some water, pour it carefully over the caustic soda and observe.

Questions:

- What is the main ingredient in the drain cleaning product and the caustic soda?
- What happened to the hair when this product was applied to it?
- What is the main ingredient of lye hair relaxers?

Conclusions:

You should have observed that the hair disintegrated when the drain cleaner was applied. The main ingredient in these cleaners is also the main ingredient in many hair relaxers, although the concentration is lower in the hair relaxers.

9.6 Chapter summary ESCPR

See presentation: 27X4 at www.everythingscience.co.za

- The **Arrhenius** definition of acids and bases defines an acid as a substance that increases the concentration of hydronium ions (H_3O^+) in a solution. A base is defined as a substance that increases the concentration of hydroxide ions (OH^-) in a solution. However, this definition only applies to aqueous solutions (in water).
- The **Brønsted-Lowry** definition is much broader. An **acid** is a substance that **donates protons** (H^+) and a **base** is a substance that **accepts protons**.
- In different reactions, certain substances can act as both an acid and a base. These substances are **amphoteric** substances. **Amphiprotic** substances are amphoteric substances that are Brønsted-Lowry acids and bases. Water is both amphoteric and amphiprotic.
- A **conjugate acid-base pair** refers to two compounds (one reactant and one product) that differ only by a hydrogen ion (H^+) and a charge of $+1$.
- A large percentage of molecules in a **strong** acid or base dissociate or ionise to

form ions in solution.

- Only a small percentage of molecules in a **weak** acid or base dissociate or ionise to form ions in solution.
- In a **concentrated** solution there is a **high ratio** of dissolved substance to solvent.
- In a **dilute** solution there is a **low ratio** of dissolved substance to solvent.
- $\mathbf{K_a}$ and $\mathbf{K_b}$ are the equilibrium constants for the reaction of an acid or a base with water. A large K_a or K_b means that the acid or base is **strong**. A small K_a or K_b means that the acid or base is **weak**.
- When an acid and a base react, they form a **salt** and **water**. The salt is made up of a cation from the base and an anion from the acid. An example of a salt is sodium chloride (NaCl), which is the product of the reaction between sodium hydroxide (NaOH) and hydrochloric acid (HCl).
- The reaction between an acid and a base is a **neutralisation** reaction.
- In the reaction between an acid and a **metal** the products are a **salt** and **hydrogen**.
- In the reaction between an acid and a **metal hydroxide** or **metal oxide** the products are a **salt** and **water**.
- In the reaction between an acid and a **metal carbontae** or **metal hydrogen carbonate** the products are a **salt**, **water** and **carbon dioxide**.
- The **pH** scale is a measure of the acidity or alkalinity of a solution. It ranges from 0 to 14. Values greater than 7 indicate a base, while those less than 7 indicate an acid.
- Water ionises to a small extent. $\mathbf{K_w}$ is a measure of this auto-ionisation. K_w is 1×10^{-14} at 25 °C.
- An **indicator** is a compound that is a different colour in a basic solution, an acidic solution, and at the end-point of a reaction. They are used to determine the end-point during a neutralisation reaction.
- **Titrations** are the method used to determine the concentration of a known substance using another, standard, solution. Acid-base titrations are an example.
- Two notable applications of acids and bases are in the **chloralkali industry**, and in hair products including **permanent waving applications**, **hair relaxers**, and **hair dyes**.

Exercise 9 – 7:

1. The stomach secretes gastric juice, which contains hydrochloric acid. The gastric juice helps with digestion. Sometimes there is an overproduction of acid, leading to heartburn or indigestion. Antacids, such as milk of magnesia, can be taken to neutralise the excess acid. Milk of magnesia is only slightly soluble in water and has the chemical formula $Mg(OH)_2$.
   a) Write a balanced chemical equation to show how the antacid reacts with the acid.
   b) The instructions on the bottle recommend that children under the age of 12 years take one teaspoon of milk of magnesia, whereas adults can take two teaspoons of the antacid. Briefly explain why the dosages are different.
   c) Why is it not advisable to take an overdose of the antacid? Refer to the hydrochloric acid concentration in the stomach in your answer.

   (DoE Grade 11 Exemplar, 2007)
2. The compound $NaHCO_3$ is commonly known as baking soda. A recipe requires 1,6 g of baking soda, mixed with other ingredients, to bake a cake.
   a) Calculate the number of moles of $NaHCO_3$ used to bake the cake.
   b) How many atoms of oxygen are there in 1,6 g of baking soda?

c) During the baking process, baking soda reacts with an acid (e.g. acetic acid in vinegar) to produce carbon dioxide and water, as shown by the reaction equation below:

$NaHCO_3(aq) + CH_3COOH(aq) \rightarrow CH_3COONa(aq) + CO_2(g) + H_2O(\ell)$

Use the above equation to explain why the cake rises during this baking process.

(DoE Grade 11 Paper 2, 2007)

3. Label the acid-base conjugate pairs in the following equation:
   $HCO_3^-(aq) + H_2O(\ell) \rightleftharpoons CO_3^{2-}(aq) + H_3O^+(aq)$
4. A certain antacid tablet contains 22,0 g of baking soda ($NaHCO_3$). It is used to neutralise the excess hydrochloric acid in the stomach. The balanced equation for the reaction is:
   $NaHCO_3(s) + HCl(aq) \rightarrow NaCl(aq) + H_2O(\ell) + CO_2(g)$
   The hydrochloric acid in the stomach has a concentration of 1,0 $mol.dm^{-3}$. Calculate the volume of the hydrochloric acid that can be neutralised by the antacid tablet.
   (DoE Grade 11 Paper 2, 2007)
5. A learner finds some sulfuric acid solution in a bottle labelled 'dilute sulfuric acid'. He wants to determine the concentration of the sulfuric acid solution. To do this, he decides to titrate the sulfuric acid against a standard potassium hydroxide (KOH) solution.
   a) What is a standard solution?
   b) Calculate the mass of KOH which he must use to make 300 cm^3 of a 0,2 $mol.dm^{-3}$ KOH solution.
   c) Calculate the pH of the 0,2 $mol.dm^{-3}$ KOH solution (assume standard temperature).
   d) Write a balanced chemical equation for the reaction between H_2SO_4 and KOH.
   e) During the titration the learner finds that 15 cm^3 of the KOH solution neutralises 20 cm^3 of the H_2SO_4 solution. Calculate the concentration of the H_2SO_4 solution.
   (IEB Paper 2, 2003)
6. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27X5   2. 27X6   3. 27X7   4. 27X8   5. 27X9

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CHAPTER 10

Electric circuits

10.1 Introduction ESCPS

The study of electrical circuits is essential to understanding the technology that uses electricity in the real-world. In this chapter we will focus on revising content from Grade 11 and extending our understanding of the internal structure of a battery/cell and how this influences what you already know about circuits.

See video: 27XB at www.everythingscience.co.za

Key linked concepts

- Units and unit conversions — Physical Sciences, Grade 10, Science skills
- Circuit components — Physical Sciences, Grade 10, Electric circuits
- Ohm's law — Physical Sciences, Grade 11, Electric circuits
- Series and parallel components — Physical Sciences, Grade 11, Electric circuits
- Equations — Mathematics, Grade 10, Equations and inequalities

10.2 Series and parallel resistor networks (Revision) ESCPT

In Grade 10 and Grade 11 you learnt about electric circuits and we introduced three quantities which are fundamental to dealing with electric circuits. These quantities are closely related and are **current, voltage (potential difference)** and **resistance**. To recap:

1. Electrical current, I, is defined as the rate of flow of charge through a circuit.
2. Potential difference or voltage, V, is related to the energy gained or lost per unit charge moving between two points in a circuit. Charge moving through a battery gains energy which is then lost moving through the circuit.
3. Resistance, R, is an internal property of a circuit element that opposes the flow of charge. Work must be done for a charge to move through a resistor.

These quantities can be related, in circuit elements whose resistance remains constant, by Ohm's law.

NOTE:
Ohm's Law
For a resistor at constant temperature the ratio $\frac{V}{I}$ is constant. This ratio we call resistance.
This is equivalent to saying that the amount of electric current through a metal conductor, at a constant temperature, in a circuit is proportional to the voltage across the conductor and can be described by

$$I = \frac{V}{R}$$

In other words, at constant temperature, the resistance of the conductor is constant, independent of the voltage applied across it or current passed through it.

We have focused on the properties of a single component. Now we need to look at a collection of components in a circuit.

IMPORTANT!

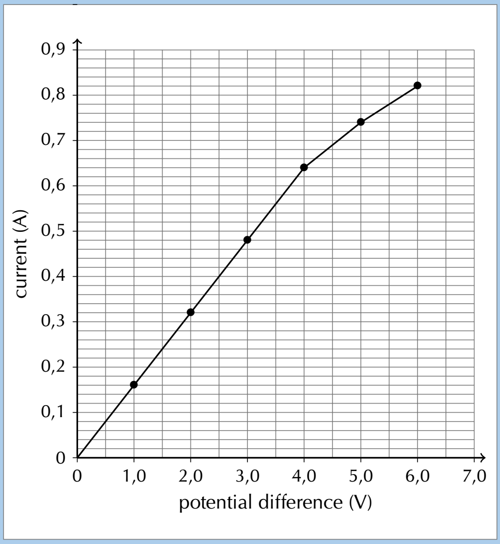
You will often hear people switch between using the terms *voltage* and *potential difference* to describe the same quantity. This is correct but very important to note.

Circuits don't consist of a single element and we've learnt about how voltage, current and resistance are affected in circuits with multiple resistors. There are two basic layouts we consider for a network of resistors, series and parallel. Resistors in series and resistors in parallel have different features when talking about current, voltage and equivalent resistance.

Worked example 1: Ohm's Law [NSC 2011 Paper 1]

QUESTION

Learners conduct an investigation to verify Ohm's law. They measure the current through a conducting wire for different potential differences across its ends. The results obtained are shown in the graph below.



1. Which ONE of the measured quantities is the dependent variable? (1 mark)
2. The graph deviates from Ohm's law at some point.
   a) Write down the coordinates of the plotted point on the graph beyond which Ohm's law is not obeyed.
   (2 marks)
   b) Give a possible reason for the deviation from Ohm's law as shown in the graph. Assume that all measurements are correct.
   (2 marks)
3. Calculate the gradient of the graph for the section where Ohm's law is obeyed.
   Use this to calculate the resistance of the conducting wire.
   (4 marks)

SOLUTION

Question 1: Current **OR** I (1 mark)

Question 2:

The graph deviates from Ohm's law at some point.

1. (4,0 ; 0,64)
   (2 marks)
2. Temperature was not kept constant.
   (2 marks)

Question 3

Gradient:
$$m = \frac{\Delta y}{\Delta x} = \frac{0,64 - 0}{4 - 0} = 0,16$$

$$R = \frac{1}{0,16} = 6,25\ \Omega$$

(4 marks)

[TOTAL: 9 marks]

IMPORTANT!

A circuit may consist of a combination of parallel and series networks that can in turn be in parallel or series. We can treat parts of the total circuit independently by applying Ohm's Law to each of the components.

NOTE:

Resistors connected in series

Resistors are in series if they are consecutive elements in the sequence of the circuit and there are no branches between them. For n resistors in series the equivalent resistance is:

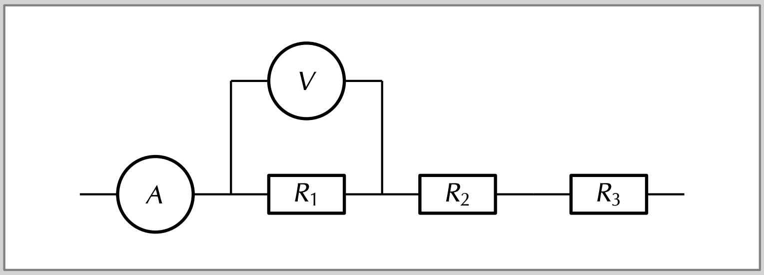
$$R_s = R_1 + R_2 + R_3 + \ldots + R_n$$

For n resistors in series the potential difference is split across the resistors:

$$V_{Total} = V_1 + V_2 + V_3 + \ldots + V_n$$

The current is constant through the resistors.

$$I_{Total} = I_1 = I_2 = I_3 = \ldots = I_n$$



It makes sense to remind ourselves of why these are consistent with other topics we have covered previously:

- **Conservation of charge:** we have learnt that charges are not created or destroyed. This is consistent with the current being the same throughout a resistor network that is in series. Charges aren't being added or lost or bunching up and, therefore, the rate at which charge moves past each point should be the same.

- **Conservation of energy:** we have learnt that energy isn't created or destroyed but transferred through work. The voltage across a resistor is the energy per unit charge (work) required to move through the resistor. The total work done to move through a network of resistors in series should be the sum of the work done to move through each individual resistor.

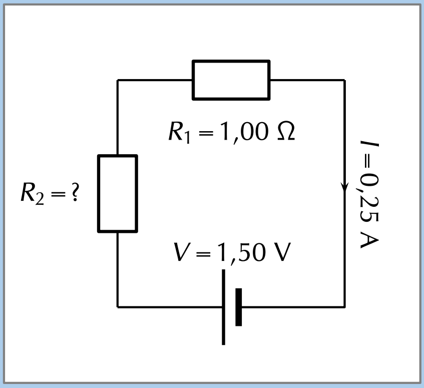
Worked example 2: Ohm's Law, all components in series

QUESTION

Two ohmic resistors (R_1 and R_2) are connected in series with a cell with negligible internal resistance. Find the resistance of R_2, given that the current flowing through R_1 and R_2 is 0,25 A and that the potential difference across the cell is 1,50 V. $R_1 = 1{,}00\ \Omega$.

SOLUTION

Step 1: Draw the circuit and fill in all known values.



Step 2: Analyse the problem.

We can use Ohm's Law to find the total resistance R in the circuit, and then calculate the unknown resistance using:

$$R = R_1 + R_2$$

because R_1 and R_2 are connected in series.

Step 3: Find the total resistance

$$\begin{aligned} R &= \frac{V}{I} \\ &= \frac{1{,}5}{0{,}25} \\ &= 6\ \Omega \end{aligned}$$

Step 4: Find the unknown resistance

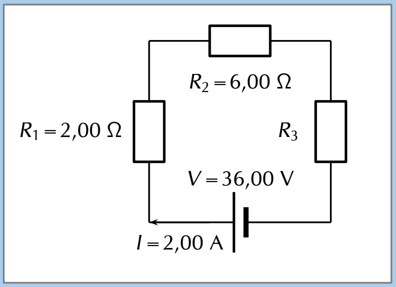
We know that $R = 6{,}00\ \Omega$ and that $R_1 = 1{,}00\ \Omega$. Since $R = R_1 + R_2$ and $R_2 = R - R_1$ we know that $R_2 = 5{,}00\ \Omega$.

Worked example 3: Ohm's Law, series circuit

QUESTION

In the case of the circuit shown, calculate:

1. the potential difference of V_1, V_2 and V_3 across the resistors R_1, R_2, and R_3
2. the resistance of R_3.



SOLUTION

Step 1: Determine how to approach the problem

We are given the potential difference across the cell and the current in the circuit, as well as the resistances of two of the three resistors. We can use Ohm's Law to calculate the potential difference across the known resistors. Since the resistors are in a series circuit the potential difference is $V = V_1 + V_2 + V_3$ and we can calculate V_3. Now we can use this information to find the potential difference across the unknown resistor R_3.

Step 2: Calculate potential difference across R_1

Using Ohm's Law:

$$\begin{aligned} R_1 &= \frac{V_1}{I} \\ I \cdot R_1 &= I \cdot \frac{V_1}{I} \\ V_1 &= I \cdot R_1 \\ &= 2 \cdot 2 \\ V_1 &= 4{,}00 \text{ V} \end{aligned}$$

Step 3: Calculate potential difference across R_2

Use Ohm's Law:

$$\begin{aligned} R_2 &= \frac{V_2}{I} \\ I \cdot R_2 &= I \cdot \frac{V_2}{I} \\ V_2 &= I \cdot R_2 \\ &= 2 \cdot 6 \\ V_2 &= 12{,}00 \text{ V} \end{aligned}$$

Step 4: Calculate potential difference across R_3

Since the potential difference across all the resistors combined must be the same as the potential difference across the cell in a series circuit, we can find V_3 using:

$$\begin{aligned} V &= V_1 + V_2 + V_3 \\ V_3 &= V - V_1 - V_2 \\ &= 36 - 4 - 12 \\ V_3 &= 20{,}00 \text{ V} \end{aligned}$$

Step 5: Find the resistance of R_3

We know the potential difference across R_3 and the current through it, so we can use Ohm's Law to calculate the value for the resistance:

$$\begin{aligned} R_3 &= \frac{V_3}{I} \\ &= \frac{20}{2} \\ R_3 &= 10{,}00\ \Omega \end{aligned}$$

Step 6: Quote the final answer

- $V_1 = 4{,}00$ V
- $V_2 = 12{,}00$ V
- $V_3 = 20{,}00$ V
- $R_3 = 10{,}00\ \Omega$

NOTE:
Equivalent resistance in a parallel network
A parallel configuration is when the current splits into a number of branches which contain components (resistors in our case). A branch may contain multiple resistors in series and still be part of the parallel configuration. For n branches of resistors in parallel, the equivalent resistance can be calculated from the total resistance of each branch and is:

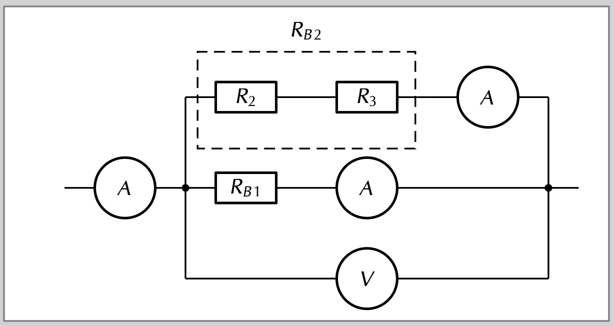
$$\frac{1}{R_p} = \frac{1}{R_{B1}} + \frac{1}{R_{B2}} + \frac{1}{R_{B3}} + \ldots + \frac{1}{R_{Bn}}$$

For n branches of resistors in parallel the potential difference is the same across each of the branches:

$$V_{Total} = V_{B1} = V_{B2} = V_{B3} = \ldots = V_{Bn}$$

The current is split through the branches.

$$I_{Total} = I_1 + I_2 + I_3 + \ldots + I_n$$



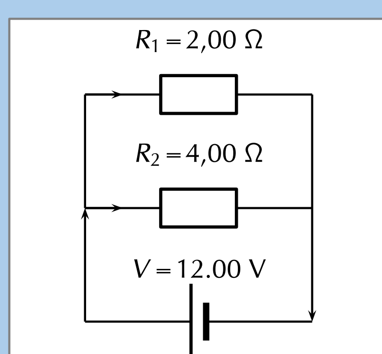
Let's take a moment to see if our conservation laws still make sense:

- **Conservation of charge:** we have learnt that charges are not created or destroyed. This is consistent with the current splitting between the branches. Charges aren't being added or lost or bunching up and, therefore, the total number of charges going through the branches must be the same as the number entering the point where the circuit branches.
- **Conservation of energy:** we have learnt that energy isn't created or destroyed but transferred through work. Energy per unit charge doesn't change unless work is done therefore it makes sense that the energy per unit charge in each branch should be the same.

Worked example 4: Ohm's Law, resistors connected in parallel

QUESTION

Calculate the current (I) in this circuit if the resistors are both ohmic in nature.



SOLUTION

Step 1: Determine what is required

We are required to calculate the total current flowing in the circuit.

Step 2: Determine how to approach the problem

Since the resistors are ohmic in nature, we can use Ohm's Law. However, there are two resistors in the circuit and we need to find the total resistance.

Step 3: Find the equivalent resistance in the circuit

Since the resistors are connected in parallel, the total (equivalent) resistance R is:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}.$$

$$\begin{aligned}\frac{1}{R} &= \frac{1}{R_1} + \frac{1}{R_2}\\ &= \frac{1}{2} + \frac{1}{4}\\ &= \frac{2+1}{4}\\ &= \frac{3}{4}\\ \text{Therefore, } R &= \frac{4}{3} = 1{,}33\ \Omega\end{aligned}$$

Step 4: Apply Ohm's Law

$$\begin{aligned}R &= \frac{V}{I}\\ R \cdot \frac{I}{R} &= \frac{V}{I} \cdot \frac{I}{R}\\ I &= \frac{V}{R}\\ I &= V \cdot \frac{1}{R}\\ &= (12)\left(\frac{3}{4}\right)\\ &= 9{,}00\ \text{A}\end{aligned}$$

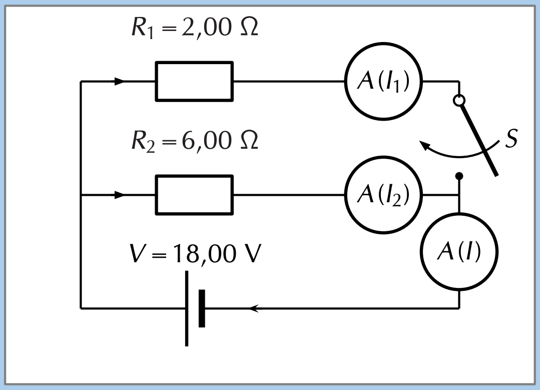
Step 5: Write the final answer

The total current flowing in the circuit is 9,00 A.

Worked example 5: Ohm's Law, parallel network of resistors

QUESTION

An 18,00 V cell is connected to two parallel resistors of 2,00 Ω and 6,00 Ω respectively. Calculate the current through each of the ammeters when the switch is closed and when it is open.



SOLUTION

Step 1: Determine how to approach the problem

We need to determine the current through the cell and each of the parallel resistors. We have been given the potential difference across the cell and the resistances of the resistors, so we can use Ohm's Law to calculate the current.

There are two alternative approaches we could adopt:

- we could use the fact that the potential difference across each of the resistors is the same as the potential difference across the battery because they are in a parallel configuration and then use Ohm's Law; or
- we could determine the equivalent resistance of the circuit and the total current and then use that to determine the current through each of the resistors.

IMPORTANT!

Both methods will result in the correct answer if you don't make any calculation errors but one is shorter.

Step 2: Now determine the current through one of the parallel resistors

We know that for a configuration with just two resistors in parallel and a cell as in this case, the potential difference across the cell is the same as the potential difference across each of the resistors in parallel. For this circuit:

$$V = V_1 = V_2 = 18{,}00 \text{ V}$$

Let's start with calculating the current through R_1 using Ohm's Law:

$$\begin{aligned} R_1 &= \frac{V_1}{I_1} \\ I_1 &= \frac{V_1}{R_1} \\ &= \frac{18{,}00}{2{,}00} \\ I_1 &= 9{,}00 \text{ A} \end{aligned}$$

Step 3: Calculate the current through the other parallel resistor

We can use Ohm's Law again to find the current in R_2:

$$\begin{aligned} R_2 &= \frac{V_2}{I_2} \\ I_2 &= \frac{V_2}{R_2} \\ &= \frac{18{,}00}{6{,}00} \\ I_2 &= 3{,}00 \text{ A} \end{aligned}$$

Step 4: Calculate the total current

The current through each of the parallel resistors must add up to the total current through the cell:

$$\begin{aligned} I &= I_1 + I_2 \\ &= 9{,}00 + 3{,}00 \\ I_2 &= 12{,}00 \text{ A} \end{aligned}$$

Step 5: When the switch is open

The branch through R_1 is not complete so no current can flow through it. This means we can ignore it completely and consider a simple circuit with a battery and a single resistor, R_2, in it.

We can use Ohm's Law again to find the current in R_2:

$$
\begin{aligned}
R_2 &= \frac{V_2}{I_2} \\
I_2 &= \frac{V_2}{R_2} \\
&= \frac{18,00}{6,00} \\
I_2 &= 3,00 \text{ A}
\end{aligned}
$$

Step 6: Write the final answer

When the switch is closed:

- The current through the cell is 12,00 A.
- The current through the 2,00 Ω resistor is 9,00 A.
- The current through the 6,00 Ω resistor is 3,00 A.

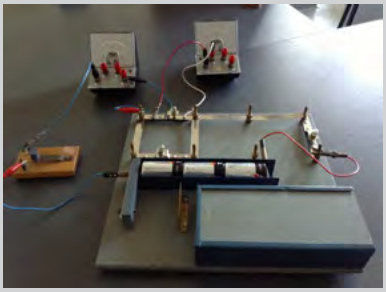
When the switch is open:

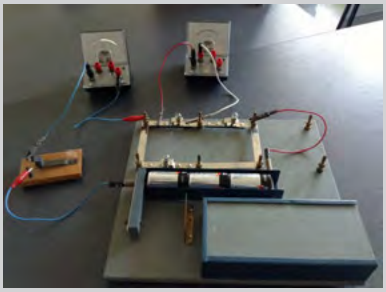
- The current through the 6,00 Ω resistor is 3,00 A.

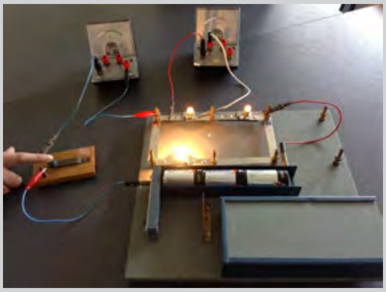
Informal experiment: Series and parallel networks

Aim:

To investigate the changes in current and voltage when branches of circuits are open or closed..







Apparatus:

You will need the following items for this investigation:

- batterys / cells
- electric leads
- a set of resistors and/or light bulbs
- ammeters
- voltmeters

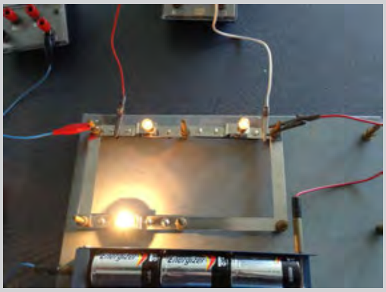
Method:

For this investigation, configure a circuit with resistors in both series and in parallel. For example, try:

- including parallel branches with different numbers of light bulbs in each branch
- changing the numbers of light bulbs or resistors in each branch
- try adding a resistor in series with the parallel network

In each branch include an ammeter and a switch. Make notes about what happens when you remove a branch by opening the switch in the branch. What happens to the current in the other branches.

Try to predict what will happen before opening or closing a switch and before adding or removing any light bulbs or resistors.

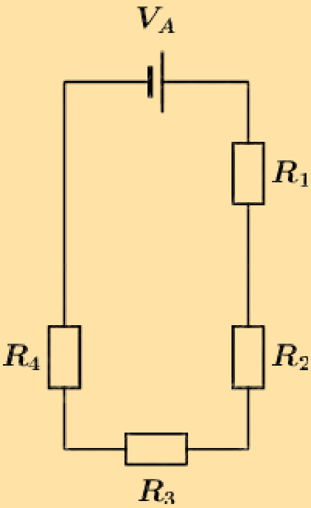


Discussion:
Be sure to note and discuss:

- whether adding a new branch increases or decreases the total current in the circuit,
- whether adding a new branch increases or decreases the current in the original branches,
- whether adding a resistor in series with a parallel network increases or decreases the current, and
- compare what happens when you add a resistor in series with adding another branch to the parallel network.

Exercise 10 – 1: Series and parallel networks

1.



The diagram shows an electric circuit consisting of a battery and four resistors. The **potential difference** (voltage) over the battery is $V_A = 2{,}8$ V
The resistors are rated as follows:

- $R_1 = 7{,}2\ \Omega$
- $R_2 = 4{,}3\ \Omega$
- $R_3 = 7{,}5\ \Omega$
- $R_4 = 4{,}1\ \Omega$

Assume that positive charge is flowing in the circuit (conventional current).
Using the concepts of **Ohm's law**, and **electric circuits**, determine the following:

a) What type of circuit is shown in the diagram?

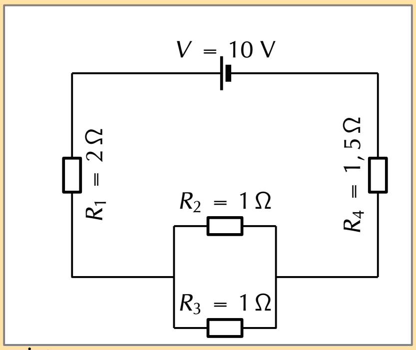
b) What is the **total equivalent resistance** R_{eq} of the circuit?
- round your answer to 1 digit after the decimal comma
- use the values for any physical constants you might need, as listed here: http://www.everythingscience.co.za/physical-constants

c) What is the potential difference (voltage) across R_1, or V_1 ?
- round your answer to 3 digits after the decimal comma
- use the values for any physical constants you might need, as listed here: http://www.everythingscience.co.za/physical-constants

d) What is the potential difference (voltage) across R_2, R_3, and R_4, or V_2, V_3, and V_4?
- round your answers to 3 digits after the decimal comma

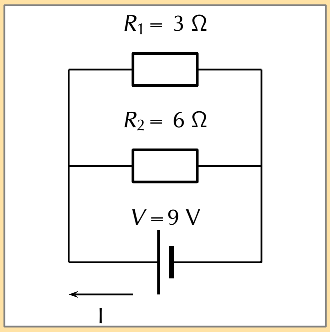
- use the values for any physical constants you might need, as listed here: http://www.everythingscience.co.za/physical-constants

2. For the following circuit, calculate:
    a) the current through the cell
    b) the potential difference across R_4
    c) the current through R_2



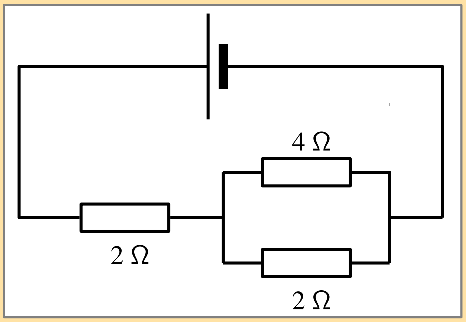
3. Calculate the equivalent resistance of:
    a) three 2 Ω resistors in series;
    b) two 4 Ω resistors in parallel;
    c) a 4 Ω resistor in series with a 8 Ω resistor;
    d) a 6 Ω resistor in series with two resistors (4 Ω and 2 Ω) in parallel.

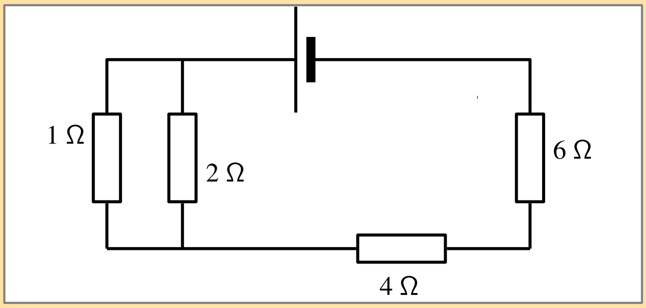
4. Calculate the total current in this circuit if both resistors are ohmic.



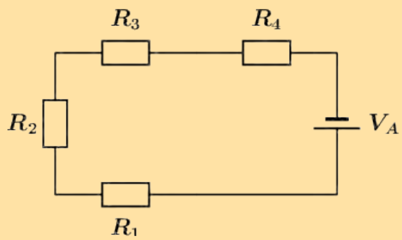
5. Two ohmic resistors are connected in series. The resistance of the one resistor is 4 Ω. What is the resistance of the other resistor if a current of 0,5 A flows through the resistors when they are connected to a voltage supply of 6 V

6. Determine the equivalent resistance of the following circuits:

6. 

7. 

The diagram shows an electric circuit consisting of a battery and four resistors. The **potential difference** (voltage) over the battery is $V_A = 1{,}2$ V



8. The the resistors are rated as follows:
   - $R_1 = 4{,}2\ \Omega$
   - $R_2 = 2{,}9\ \Omega$
   - $R_3 = 3{,}8\ \Omega$
   - $R_4 = 3{,}5\ \Omega$

   Assume that positive charge is flowing in the circuit (conventional current).

   Using the concepts of **Ohm's law**, and **electric circuits**, determine the following:

   a) What type of circuit is shown in the diagram?
   b) What is the **total equivalent resistance** R_{eq} of the circuit?
      - round your answer to 1 digit after the decimal comma
      - use the values for any physical constants you might need, as listed here: http://www.everythingscience.co.za/physical-constants
   c) What is the potential difference (voltage) across R_1, or V_1 ?
      - round your answer to 3 digits after the decimal comma
      - use the values for any physical constants you might need, as listed here: http://www.everythingscience.co.za/physical-constants
   d) What is the potential difference (voltage) across R_2, R_3, and R_4, or V_2, V_3, and V_4?
      - round your answers to 3 digits after the decimal comma
      - use the values for any physical constants you might need, as listed here: http://www.everythingscience.co.za/physical-constants

9. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27XC 2. 27XD 3. 27XF 4. 27XG 5. 27XH 6. 27XJ
7. 27XK 8. 27XM

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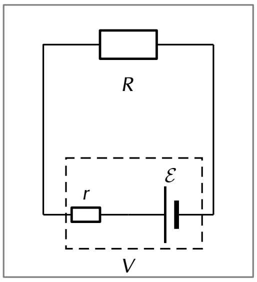
10.3 Batteries and internal resistance ESCPV

Up until now we have been dealing with ideal batteries in that they aren't affected by the circuit or current in any way and provide a precise voltage until they go flat.

If you measure the potential difference across the terminals of a battery on its own you will get a different value to what you measure when it is in a complete circuit. The value will be less when the battery is included in a complete circuit. Sometimes the difference is called the *lost volts*. Nothing has actually been lost but energy has been transferred.

Real batteries are made from materials which have resistance. This means that real batteries are not just sources of potential difference (voltage), but they also possess internal resistance. If the total potential difference source is referred to as the emf, $\mathcal{E}$, then a real battery can be represented as an emf connected in series with a resistor r. The internal resistance of the battery is represented by the symbol r.



DEFINITION: *Load*

The external resistance in the circuit is referred to as the load.

Suppose that the battery with emf $\mathcal{E}$ and internal resistance r supplies a current I through an external load resistor R. Then the potential difference across the load resistor is that supplied by the battery: $V_{\text{load}} = I \cdot R$

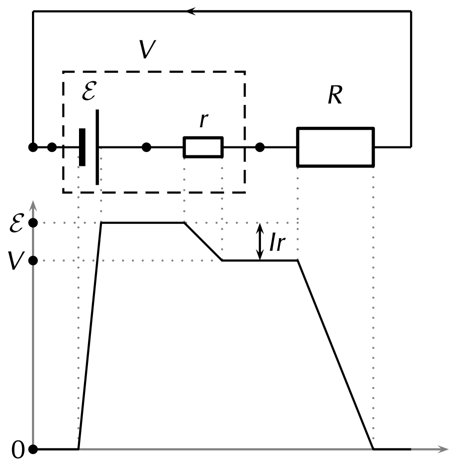
Similarly, from Ohm's Law, the potential difference across the internal resistance is: $V_{\text{internal resistance}} = I \cdot r$
The potential difference V of the battery is related to its emf $\mathcal{E}$ and internal resistance r by:

$$\mathcal{E} = V + Ir$$

or

$$V = \mathcal{E} - Ir$$

The battery is the source of energy and the energy provided per unit charge (emf) passing through the battery is equal to the total work done (potential difference) across the components in the circuit. This can be illustrated by showing the energy per unit charge as as function of the position in the circuit. The charge gains energy when moving through the battery and loses energy when moving through resistors.



1. Postive work is done on a unit charge by the battery transferring an energy equal to the emf.
2. The charge does work to overcome the internal resistance of the battery. Doing work requires that the charge lose some energy. The work done to overcome the internal resistance is $V_{\text{internal resistance}} = Ir$.
3. When the unit charge leaves the battery it has less energy than the original emf. This is now the total energy that it can use to do work moving through the circuit, $V_{\text{load}} = \mathcal{E} - Ir$.
4. As the charge moves around the circuit it does work passing through each component which is equal to IR.

The emf of a battery is essentially constant because it only depends on the chemical reaction (that converts chemical energy into electrical energy) going on inside the battery. Therefore, we can see that the potential difference across the terminals of the battery is dependent on the current drawn by the load. The higher the current, the lower the potential difference across the terminals, because the emf is constant. For the same reason, the potential difference only equals the emf when the current is very small.

The current that can be drawn from a battery is limited by a critical, maximum value I_c. The more resistance in the circuit the less the current will be. Imagine that you have a wire with no resistance that you use to connect the terminals of the battery. The circuit is complete, current will flow and adding any resistance to the circuit would decrease the current. The current without any external resistance will be I_c. At a current of I_c, $V = 0$ V because there is no load in the circuit. Then, the equation becomes:

$$\begin{aligned} 0 &= \mathcal{E} - I_c r \\ I_c r &= \mathcal{E} \\ I_c &= \frac{\mathcal{E}}{r} \end{aligned}$$

The maximum current that can be drawn from a battery is less than $\frac{\mathcal{E}}{r}$.

Worked example 6: Internal resistance

QUESTION

Determine the internal resistance of a battery that has an emf of 12,00 V and has a potential difference across its terminals of 10,00 V when a current of 4,00 A is flowing through the battery when connected in a circuit.

SOLUTION

Step 1: Determine how to approach the problem
It is an internal resistance problem. So we use the equation:

$$\mathcal{E} = V + Ir$$

Step 2: Solve the problem

$$\begin{aligned} \mathcal{E} &= V + Ir \\ 12 &= 10 + 4(r) \\ r &= 0{,}5\ \Omega \end{aligned}$$

Step 3: Write the final answer
The internal resistance of the battery is 0,50 Ω.

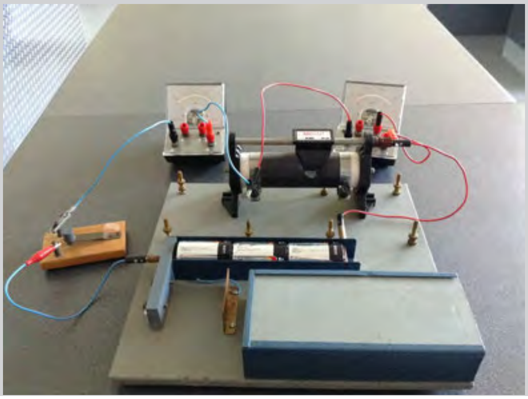
Formal experiment: Finding the internal resistance of a battery

Aim:

To determine the internal resistance of a battery.

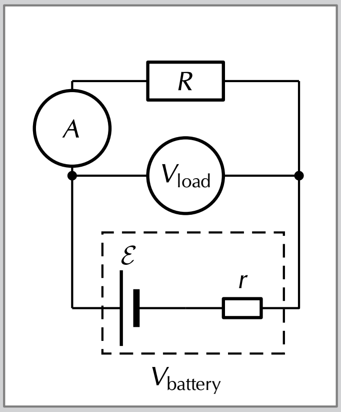
Apparatus:
You will need the following items for this experiment:

- battery / cell to be studied
- electric leads
- a set of resistors or a rheostat
- ammeter
- voltmeter



Method:

We will set up a circuit that contains the battery we want to study in series with a resistor. Then we will measure the potential difference across the load as well as the current for a number of different resistors/rheostat in the circuit. It doesn't matter if it is a different resistor each time or more resistors in series or parallel. What matters is that the overall resistance of the circuit changes so that the current is different each time. The reason that doing this can tell us about the internal resistance of the battery is because the potential difference across the internal resistance is $V_{\text{internal resistance}} = I \cdot r$ and we can vary I by changing the resistance of the circuit.



If the potential difference across the internal resistance is changing and we add up all the potential differences, $\mathcal{E} = V_{\text{load}} + V_{\text{internal resistance}}$ we can determine the internal resistance.

To do this we will actually plot a graph of V_{load} versus I and then use the features of the graph to determine $\mathcal{E}$ and r. To understand why plotting the graph will help us we start with the equation for the magnitude of the $\mathcal{E}$ and substitute Ohm's Law and re-arrange as follows:

$$\begin{aligned}
\mathcal{E} &= V_{\text{load}} + V_{\text{internal resistance}} \\
\mathcal{E} &= V_{\text{load}} + I \cdot r \\
V_{\text{load}} &= \mathcal{E} - I \cdot r \\
V_{\text{load}} &= -r \cdot I + \mathcal{E} \\
\underbrace{V_{\text{load}}}_{y} &= \underbrace{-r}_{m} \cdot \underbrace{I}_{x} + \underbrace{\mathcal{E}}_{c}
\end{aligned}$$

If we plot V_{load} versus I we will be plotting data that are governed by this relationship. This allows us to conclude that the slope of the graph, m, will be $-r$ and the intercept vertical axis, c, will be the emf $\mathcal{E}$.

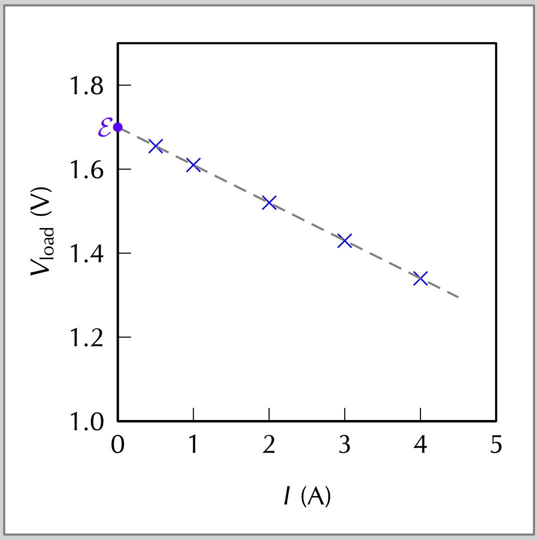
Results:

Record your results in a table like the one alongside. You can take more readings if you like.

Setup	V_{load} (V)	I (A)
Resistance 1		
Resistance 2		
Resistance 3		
Resistance 4		
Resistance 5		

Discussion and conclusion:

Plot your data on a set of axes similar to this example. The blue crosses represent the measured data points, the gray, dashed line is the drawn straight line through the data points. The best fit line you draw doesn't need to go through all the data points, it should, in general, have as many points above and below the line. The slope of the line can be measured and equated to $-r$ and the intercept with the vertical axis will give you $\mathcal{E}$. The intercept with the horizontal axis would give you the maximum possible current the battery could deliver.



- Do your data form a perfectly straight line?
- What errors were introduced?

Exercise 10 – 2:

1. Describe what is meant by the *internal resistance* of a real battery.
2. Explain why there is a difference between the emf and terminal voltage of a battery if the load (external resistance in the circuit) is comparable in size to the battery's internal resistance
3. What is the internal resistance of a battery if its emf is 6 V and the potential difference across its terminals is 5,8 V when a current of 0,5 A flows in the circuit when it is connected across a load?
4. A 12,0 V battery has an internal resistance of 7,0 Ω.
   a) What is the maximum current this battery could supply?
   b) What is the potential difference across its terminals when it is supplying a current of 150.0 mA?
   c) Draw a sketch graph to show how the terminal potential difference varies with the current supplied if the internal resistance remains constant. How could the internal resistance be obtained from the graph?
5. In a hearing aid a battery supplies a current of 25.0 mA through a resistance of 400 Ω. When the volume is increased, the resistance is changed to 100 Ω and the current rises to 60 mA. What is the emf and internal resistance of the cell?
6. A battery is connected in series with a rheostat and an ammeter. When the resistance of the resistor is 10 Ω the current is 2.0 A. When the resistance is 5 Ω the current is 3.8 A. Find the emf and the internal resistance of the battery.
7. When a cell is connected directly across a high resistance voltmeter the reading is 1.50 V. When the cell is shorted through a low resistance ammeter the current

is 2.5 A. What is the emf and internal resistance of the cell?

8. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27XN 2. 27XP 3. 27XQ 4. 27XR 5. 27XS 6. 27XT
7. 27XV

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10.4 Evaluating internal resistance in circuits ESCPW

Approach ESCPX

The approach to solving problems that involve the internal resistance of batteries is straightforward, you just need to understand that each battery in previous examples was a source of emf, $\mathcal{E}$, and a small resistor, r, and then solve as before but include r in your calculations.

An important thing to realise is that the potential difference you calculated or were given in previous examples is not the emf, it is the emf less that potential difference across the internal resistance.

To emphasise that internal resistance is an extension to what you have already done we are going to take previous worked examples and consider the internal resistance of the battery. If the internal resistance did not behave like an ohmic resistor this wouldn't be possible but we won't deal with cases like that.

Applications ESCPY

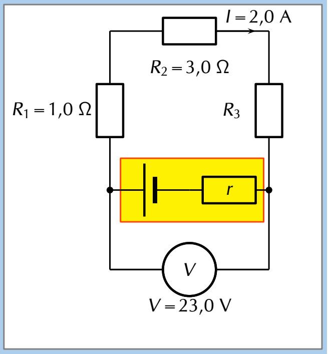
Worked example 7: Internal resistance in circuit with resistors in series

QUESTION

For the circuit shown, calculate:

1. the potential differences V_1, V_2 and V_3 across the resistors R_1, R_2, and R_3 .
2. the resistance of R_3.
3. the resistance of R_3.

If the internal resistance is 0,1 Ω, what is the emf of the battery and what power is dissipated by the internal resistance of the battery?



SOLUTION

Step 1: Note

This is a very similar question to what you have seen earlier. This is to highlight the fact that the approach when dealing with internal resistance is built on all the same principles you have already been working with.

Step 2: Determine how to approach the problem

We are given the potential difference across the cell and the current in the circuit, as well as the resistances of two of the three resistors. We can use Ohm's Law to calculate the potential difference across the known resistors. Since the resistors are in a series circuit the potential difference is $V = V_1 + V_2 + V_3$ and we can calculate V_3. Now we can use this information to find the potential difference across the unknown resistor R_3.

Step 3: Calculate potential difference across R_1

Using Ohm's Law:

$$\begin{aligned} R_1 &= \frac{V_1}{I} \\ I \cdot R_1 &= I \cdot \frac{V_1}{I} \\ V_1 &= I \cdot R_1 \\ &= 2 \cdot 1 \\ V_1 &= 2 \text{ V} \end{aligned}$$

Step 4: Calculate potential difference across R_2

Again using Ohm's Law:

$$\begin{aligned} R_2 &= \frac{V_2}{I} \\ I \cdot R_2 &= I \cdot \frac{V_2}{I} \\ V_2 &= I \cdot R_2 \\ &= 2 \cdot 3 \\ V_2 &= 6 \text{ V} \end{aligned}$$

Step 5: Calculate potential difference across R_3

Since the potential difference across all the resistors combined must be the same as the potential difference across the cell in a series circuit, we can find V_3 using:

$$\begin{aligned} V &= V_1 + V_2 + V_3 \\ V_3 &= V - V_1 - V_2 \\ &= 23 - 2 - 6 \\ V_3 &= 15 \text{ V} \end{aligned}$$

Step 6: Find the resistance of R_3

We know the potential difference across R_3 and the current through it, so we can use Ohm's Law to calculate the value for the resistance:

$$\begin{aligned} R_3 &= \frac{V_3}{I} \\ &= \frac{15}{2} \\ R_3 &= 7{,}5\ \Omega \end{aligned}$$

Step 7: Potential difference across the internal resistance of the battery

The value of the emf can be calculated from the potential difference of the load and the potential difference across the internal resistance.

$$\begin{aligned} \mathcal{E} &= V + Ir \\ &= 23 + (2)(0{,}1) \\ &= 23{,}2 \text{ V} \end{aligned}$$

Step 8: Power dissipated in the battery

We know that the power dissipated in a resistor is given by $P = VI = I^2R = \frac{V^2}{R}$ and we know the current in the circuit, the internal resistance and the potential difference across it so we can use any form of the equation for power:

$$\begin{aligned} P_r &= V_r I_r \\ &= (0{,}2)(2) \\ &= 0{,}4 \text{ W} \end{aligned}$$

Step 9: Write the final answer

- $V_1 = 2{,}0$ V
- $V_2 = 6{,}0$ V
- $V_3 = 10{,}0$ V
- $R_3 = 7{,}5\Omega$
- $\mathcal{E} = 23{,}2$ V
- $P_r = 0{,}4$ W

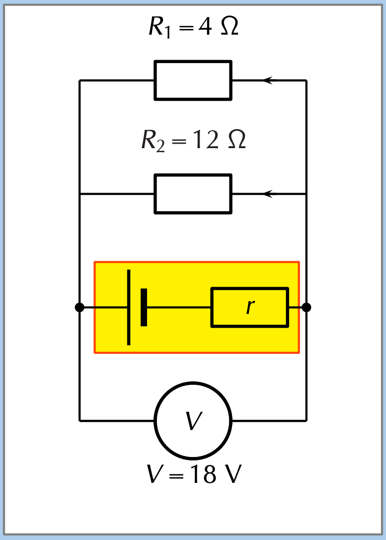
Worked example 8: Internal resistance and resistors in parallel

QUESTION

The potential difference across a battery measures 18 V when it is connected to two parallel resistors of 4,00 Ω and 12,00 Ω respectively. Calculate the current through the cell and through each of the resistors. If the internal resistance of the battery is 0,375 Ω what is the emf of the battery?

SOLUTION

Step 1: First draw the circuit before doing any calculations



Step 2: Determine how to approach the problem

We need to determine the current through the cell and each of the parallel resistors. We have been given the potential difference across the cell and the resistances of the resistors, so we can use Ohm's Law to calculate the current.

Step 3: Calculate the current through the cell

To calculate the current through the cell we first need to determine the equivalent resistance of the rest of the circuit. The resistors are in parallel and therefore:

$$\begin{aligned}\frac{1}{R} &= \frac{1}{R_1} + \frac{1}{R_2}\\ &= \frac{1}{4} + \frac{1}{12}\\ &= \frac{3+1}{12}\\ &= \frac{4}{12}\\ R &= \frac{12}{4} = 3{,}00\ \Omega\end{aligned}$$

Now using Ohm's Law to find the current through the cell:

$$\begin{aligned} R &= \frac{V}{I} \\ I &= \frac{V}{R} \\ &= \frac{18}{3} \\ I &= 6{,}00 \text{ A} \end{aligned}$$

Step 4: Now determine the current through one of the parallel resistors

We know that for a purely parallel resistor configuration, the potential difference across the cell is the same as the potential difference across each of the parallel resistors. For this circuit:

$$V = V_1 = V_2 = 18 \text{ V}$$

Let's start with calculating the current through R_1 using Ohm's Law:

$$\begin{aligned} R_1 &= \frac{V_1}{I_1} \\ I_1 &= \frac{V_1}{R_1} \\ &= \frac{18}{4} \\ I_1 &= 4{,}50 \text{ A} \end{aligned}$$

Step 5: Calculate the current through the other parallel resistor

We can use Ohm's Law again to find the current in R_2:

$$\begin{aligned} R_2 &= \frac{V_2}{I_2} \\ I_2 &= \frac{V_2}{R_2} \\ &= \frac{18}{12} \\ I_2 &= 1{,}50 \text{ A} \end{aligned}$$

An alternative method of calculating I_2 would have been to use the fact that the currents through each of the parallel resistors must add up to the total current through the cell:

$$\begin{aligned} I &= I_1 + I_2 \\ I_2 &= I - I_1 \\ &= 6 - 4.5 \\ I_2 &= 1{,}5 \text{ A} \end{aligned}$$

Step 6: Determine the emf

This total current through the battery is the current through the internal resistance of the battery. Knowing the current and resistance allows us to use Ohm's law to determine the potential difference across the internal resistance and therefore the emf of the battery.

Using Ohm's law we can determine the potential difference across the internal resistance:

$$\begin{aligned} V &= I \cdot r \\ &= 6 \cdot 0{,}375 \\ &= 2{,}25 \text{ V} \end{aligned}$$

We know that the emf of the battery is the potential difference across the terminal summed with the potential difference across the internal resistance so:

$$\begin{aligned} \mathcal{E} &= V + Ir \\ &= 18 + 2{,}25 \\ &= 20{,}25 \text{ V} \end{aligned}$$

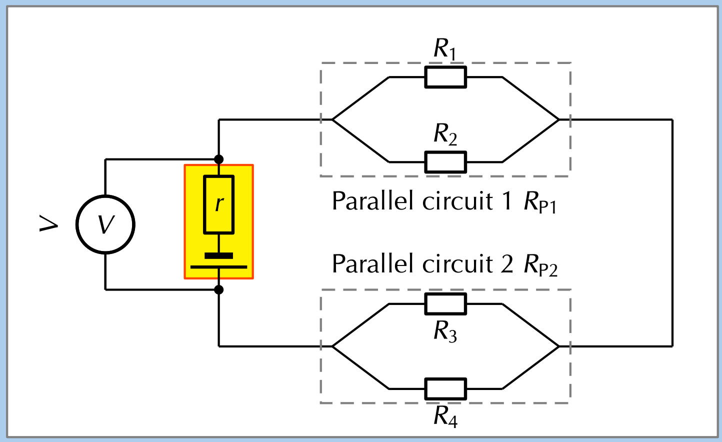
Step 7: Write the final answer

The current through the cell is 6,00 A. The current through the 4,00 Ω resistor is 4,50 A. The current through the 12,00 Ω resistor is 1,50 A. The emf of the battery is 20,25 V.

Worked example 9: Power in series and parallel networks of resistors

QUESTION

Given the following circuit:



The current leaving the battery is 1,07 A, the total power dissipated in the external circuit is 6,42 W, the ratio of the total resistances of the two parallel networks $R_{P1} : R_{P2}$ is 1:2, the ratio $R_1 : R_2$ is 3:5 and $R_3 = 7{,}00\ \Omega$.

Determine the:

1. potential difference of the battery,
2. the power dissipated in R_{P1} and R_{P2}, and
3. if the battery is labelled as having an emf of 6,50 V what is the value of the resistance of each resistor and the power dissipated in each of them.

SOLUTION

Step 1: What is required

In this question you are given various pieces of information and asked to determine the power dissipated in each resistor and each combination of resistors. Notice that the information given is mostly for the overall circuit. This is a clue that you should start with the overall circuit and work downwards to more specific circuit elements.

Step 2: Calculating the potential difference of the battery

Firstly we focus on the battery. We are given the power for the overall circuit as well as the current leaving the battery. We know that the potential difference across the terminals of the battery is the potential difference across the circuit as a whole.

We can use the relationship $P = VI$ for the entire circuit because the potential difference is the same as the potential difference across the terminals of the battery:

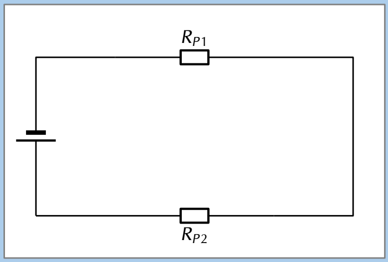
$$\begin{aligned} P &= VI \\ V &= \frac{P}{I} \\ &= \frac{6{,}42}{1{,}07} \\ &= 6{,}00\ \text{V} \end{aligned}$$

The potential difference across the battery is 6,00 V.

Step 3: Power dissipated in R_{P1} and R_{P2}

Remember that we are working from the overall circuit details down towards those for individual elements, this is opposite to how you treated this circuit earlier.

We can treat the parallel networks like the equivalent resistors so the circuit we are currently dealing with looks like:



We know that the current through the two circuit elements will be the same because it is a series circuit and that the resistance for the total circuit must be: $R_{Ext} = R_{P1} + R_{P2}$. We can determine the total resistance from Ohm's Law for the circuit as a whole:

$$\begin{aligned} V_{battery} &= IR_{Ext} \\ R_{Ext} &= \frac{V_{battery}}{I} \\ &= \frac{6,00}{1,07} \\ &= 5,61\ \Omega \end{aligned}$$

We know that the ratio between $R_{P1} : R_{P2}$ is 1:2 which means that we know:

$$\begin{aligned} R_{P1} &= \frac{1}{2}R_{P2} \text{ and} \\ R_T &= R_{P1} + R_{P2} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2}R_{P2} + R_{P2} \\ &= \frac{3}{2}R_{P2} \\ (5,61) &= \frac{3}{2}R_{P2} \\ R_{P2} &= \frac{2}{3}(5,61) \\ R_{P2} &= 3,74\ \Omega \end{aligned}$$

and therefore:

$$\begin{aligned} R_{P1} &= \frac{1}{2}R_{P2} \\ &= \frac{1}{2}(3.74) \\ &= 1,87\ \Omega \end{aligned}$$

Now that we know the total resistance of each of the parallel networks we can calculate the power dissipated in each:

$$\begin{aligned} P_{P1} &= I^2R_{P1} \\ &= (1,07)^2(1,87) \\ &= 2,14\ \text{W} \end{aligned}$$

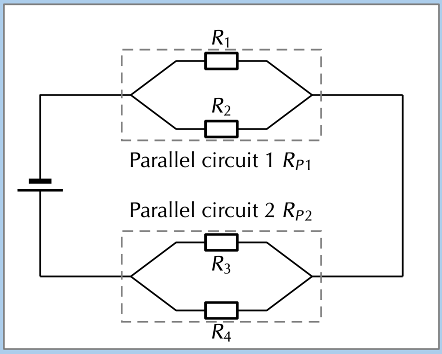
and

$$\begin{aligned} P_{P2} &= I^2R_{P2} \\ &= (1,07)^2(3,74) \\ &= 4,28\ \text{W} \end{aligned}$$

These values will add up to the original power value we had for the external circuit. If they didn't we would have made a calculation error.

Step 4: Parallel network 1 calculations

Now we can begin to do the detailed calculation for the first set of parallel resistors.



We know that the ratio between $R_1 : R_2$ is 3:5 which means that we know $R_1 = \frac{3}{5}R_2$. We also know the total resistance for the two parallel resistors in this network is 1,87 Ω. We can use the relationship between the values of the two resistors as well as the formula for the total resistance ($\frac{1}{R_PT} = \frac{1}{R_1} + \frac{1}{R_2}$) to find the resistor values:

$$\frac{1}{R_{P1}} = \frac{1}{R_1} + \frac{1}{R_2}$$
$$\frac{1}{R_{P1}} = \frac{5}{3R_2} + \frac{1}{R_2}$$
$$\frac{1}{R_{P1}} = \frac{1}{R_2}(\frac{5}{3} + 1)$$
$$\frac{1}{R_{P1}} = \frac{1}{R_2}(\frac{5}{3} + \frac{3}{3})$$
$$\frac{1}{R_{P1}} = \frac{1}{R_2}\frac{8}{3}$$
$$R_2 = R_{P1}\frac{8}{3}$$
$$= (1,87)\frac{8}{3}$$
$$= 4,99\ \Omega$$

We can also calculate R_1:

$$R_1 = \frac{3}{5}R_2$$
$$= \frac{3}{5}(4,99)$$
$$= 2,99\ \Omega$$

To determine the power we need the resistance which we have calculated and either the potential difference or current. The two resistors are in parallel so the potential difference across them is the same as well as the same as the potential difference across the parallel network. We can use Ohm's Law to determine the potential difference across the network of parallel resistors as we know the total resistance and we know the current:

$$V = IR$$
$$= (1,07)(1,87)$$
$$= 2,00\ \text{V}$$

We now have the information we need to determine the power through each resistor:

$$P_1 = \frac{V^2}{R_1}$$
$$= \frac{(2,00)^2}{2,99}$$
$$= 1,34\ \text{W}$$

$$P_2 = \frac{V^2}{R_2}$$
$$= \frac{(2,00)^2}{4,99}$$
$$= 0,80\ \text{W}$$

Step 5: Parallel network 2 calculations

Now we can begin to do the detailed calculation for the second set of parallel resistors.
We are given $R_3 = 7,00\ \Omega$ and we know R_{P2} so we can calculate R_4 from:

$$\frac{1}{R_{P2}} = \frac{1}{R_3} + \frac{1}{R_4}$$
$$\frac{1}{3,74} = \frac{1}{7,00} + \frac{1}{R_4}$$
$$R_4 = 8,03\ \Omega$$

We can calculate the potential difference across the second parallel network by subtracting the potential difference of the first parallel network from the battery potential difference, $V_{P2} = 6,00 - 2,00 = 4,00$ V.

We can now determine the power dissipated in each resistor:

$$P_3 = \frac{V^2}{R_3}$$
$$= \frac{(4,00)^2}{7,00}$$
$$= 2,29\ \text{W}$$

$$P_4 = \frac{V^2}{R_2}$$
$$= \frac{(4,00)^2}{8,03}$$
$$= 1,99\ \text{W}$$

Step 6: Internal resistance

We know that the emf of the battery is 6,5 V but that the potential difference measured across the terminals is only 6 V.

The difference is the potential difference across the internal resistance of the battery and we can use the known current and Ohm's law to determine the internal resistance:

$$V = I \cdot R$$
$$R = \frac{V}{I}$$
$$= \frac{0,5}{1,07}$$
$$= 0,4672897$$
$$= 0,47\ \Omega$$

The power dissipated by the internal resistance of the battery is:

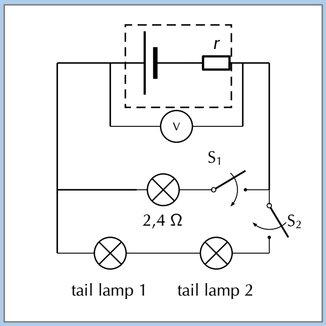
$$P = VI$$
$$= 0,5 \cdot 1,07$$
$$= 0,535\ \text{W}$$

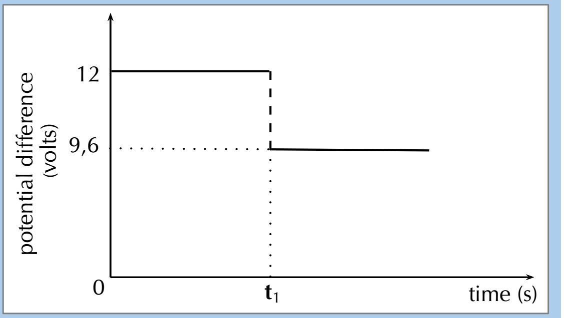
Worked example 10: Internal resistance and headlamps [NSC 2011 Paper 1]

QUESTION

The headlamp and two IDENTICAL tail lamps of a scooter are connected in parallel to a battery with unknown internal resistance as shown in the simplified circuit diagram below. The headlamp has a resistance of 2,4 Ω and is controlled by switch $\mathbf{S}_1$. The tail lamps are controlled by switch $\mathbf{S}_2$. The resistance of the connecting wires may be ignored.

The graph alongside shows the potential difference across the terminals of the battery before and after switch $\mathbf{S}_1$ is closed (whilst switch $\mathbf{S}_2$ is open). Switch $\mathbf{S}_1$ is closed at time $\mathbf{t}_1$.





1. Use the graph to determine the emf of the battery. (1 mark)
2. WITH ONLY SWITCH $\mathbf{S}_1$ CLOSED, calculate the following:
   a) Current through the headlamp (3 marks)
   b) Internal resistance, *r*, of the battery (3 marks)
3. BOTH SWITCHES $\mathbf{S}_1$ AND $\mathbf{S}_2$ ARE NOW CLOSED. The battery delivers a current of 6 A during this period.
   Calculate the resistance of each tail lamp. (5 marks)
4. How will the reading on the voltmeter be affected if the headlamp burns out? (Both switches $\mathbf{S}_1$ and $\mathbf{S}_2$ are still closed.)

Write down only INCREASES, DECREASES or REMAINS THE SAME.
Give an explanation. (3 marks)

SOLUTION

Question 1: 12 V (1 mark)

Question 2.1

Option 1:

$$I = \frac{V}{R} = \frac{9,6}{2,4} = 4 \text{ A}$$

Option 2:

$$\text{emf} = IR + Ir$$
$$12 = I(2,4) + 2,4$$
$$\therefore I = 4 \text{ A}$$

(3 marks)

Question 2.2

Option 1:

$$\text{emf} = IR + Ir$$
$$12 = 9,4 + 4r$$
$$r = 0,6\ \Omega$$

Option 2:

$$V_{lost} = Ir$$
$$2,4 = 4r$$
$$\therefore r = 0,6\ \Omega$$

Option 3:

$$\text{emf} = I(R + r)$$
$$12 = 4(2,4 + r)$$
$$\therefore r = 0,6\ \Omega$$

(3 marks)

Question 3

Option 1:

$$\text{emf} = IR + Ir$$
$$12 = 6(R + 0,6)$$
$$R_{\text{ext}} = 1,4\ \Omega$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$
$$\frac{1}{1,4} = \frac{1}{2,4} + \frac{1}{R}$$
$$R = 3,36\ \Omega$$

Each tail lamp: $R = 1,68\ \Omega$

Option 2:

$$\text{Emf} = V_{\text{terminal}} + Ir$$
$$12 = V_{\text{terminal}} + 6(0,6)$$
$$\therefore V_{\text{terminal}} = 8,4 \text{ V}$$

$$I_{2,4\ \Omega} = \frac{V}{R} = \frac{8,4}{2,4} = 3,5 \text{ A}$$

$$I_{\text{tail lamps}} = 6 - 3,5 = 2,5 \text{ A}$$

$$R_{\text{tail lamps}} = \frac{V}{I} = \frac{8,4}{2,5} = 3,36\ \Omega$$

$$R_{\text{tail lamp}} = 1,68\ \Omega$$

Option 3:

$$V = IR$$
$$12 = 6(R)$$
$$R_{ext} = 2\ \Omega$$

$$R_{parallel} = 2 - 0{,}6$$
$$= 1{,}4\ \Omega$$
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$
$$\frac{1}{1{,}4} = \frac{1}{2{,}4} + \frac{1}{R}$$
$$R = 3{,}36\ \Omega$$

Each tail lamp: $R = 1{,}68\ \Omega$

Option 4:

For parallel combination: $I_1 + I_2 = 6$ A

$$\therefore \frac{V}{2{,}4} + \frac{V}{R_{tail\ lamps}} = 6$$
$$8{,}4\left(\frac{1}{2{,}4} + \frac{1}{R_{tail\ lamps}}\right) = 6$$
$$\therefore R_{tail\ lamps} = 3{,}36\ \Omega$$
$$R_{tail\ lamp} = 1{,}68\ \Omega$$

(5 marks)

Question 4

Increases. The resistance increases and the current decreases. So Ir (lost volts) must decrease which leads to an increase in the voltage. (3 marks)

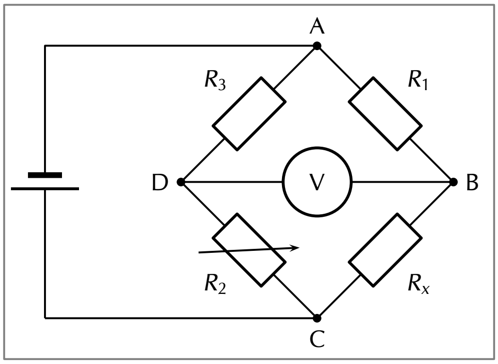
[TOTAL: 15 marks]

FACT
The Wheatstone bridge was invented by Samuel Hunter Christie in 1833 and improved and popularised by Sir Charles Wheatstone in 1843.

10.5 Extension: Wheatstone bridge [Not examinable] ES-CPZ

Using what we know about parallel networks of resistors we can devise another method of finding an unknown resistance, the *Wheatstone bridge*. A Wheatstone bridge is a measuring instrument that is used to measure an unknown electrical resistance by balancing two legs of a bridge circuit, one leg of which includes the unknown component. Its operation is similar to the original potentiometer except that in potentiometer circuits the meter used is a sensitive galvanometer.

Despite the fact that we have given this circuit a special name, it is just a circuit containing a parallel configuration of four resistors. This is not actually a new concept, this particular configuration is just particularly useful.

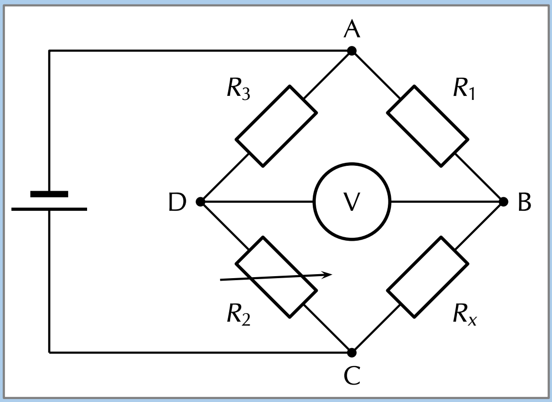


In the circuit of the Wheatstone bridge, R_x is the unknown resistance. R_1, R_2 and R_3 are resistors of known resistance and the resistance of R_2 is adjustable. If the ratio of R_2: R_1 is equal to the ratio of R_x: R_3, then the potential difference between the two midpoints will be zero and no current will flow between the midpoints. In order to determine the unknown resistance, R_2 is varied until this condition is reached. That is when the voltmeter reads 0 V.

Worked example 11: Wheatstone bridge

QUESTION

What is the resistance of the unknown resistor R_x in the diagram below if $R_1 = 4\ \Omega$ $R_2 = 8\ \Omega$ and $R_3 = 6\ \Omega$.



SOLUTION

Step 1: Determine how to approach the problem

The arrangement is a Wheatstone bridge. So we use the equation:

$$R_x : R_3 = R_2 : R_1$$

Step 2: Solve the problem

$$\begin{aligned} R_x : R_3 &= R_2 : R_1 \\ R_x : 6 &= 8 : 4 \\ R_x &= 12\ \Omega \end{aligned}$$

Step 3: Write the final answer

The resistance of the unknown resistor is 12 Ω.

10.6 Chapter summary ESCQ2

See presentation: 27XW at www.everythingscience.co.za

1. Ohm's Law governs the relationship between current and potential difference for a circuit element at constant temperature. Mathematically we write $I = \frac{V}{R}$.
2. Conductors that obey Ohm's Law are called ohmic conductors; those that do not are called non-ohmic conductors.
3. Ohm's Law can be applied to a single circuit element or the circuit as a whole (if the components are ohmic).
4. The equivalent resistance of resistors in series (R_s) can be calculated as follows: $R_s = R_1 + R_2 + R_3 + \ldots + R_n$
5. The equivalent resistance of resistors in parallel (R_p) can be calculated as follows: $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots + \frac{1}{R_n}$
6. Real batteries have an internal resistance.
7. The potential difference V of the battery is related to its emf $\mathcal{E}$ and internal resistance r by:

$$\mathcal{E} = V_{\text{load}} + V_{\text{internal resistance}}$$

or

$$\mathcal{E} = IR_{Ext} + Ir$$

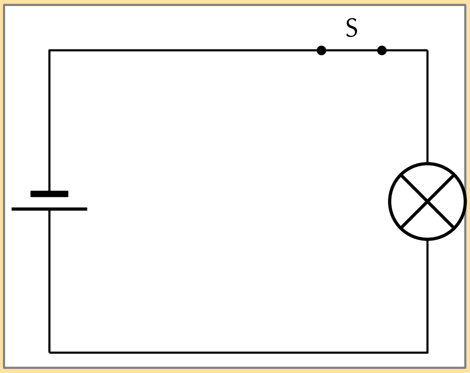
8. The external resistance in the circuit is referred to as the load.

Physical Quantities		
Quantity	**Unit name**	**Unit symbol**
Current (I)	Amperes	A
Electrical energy (E)	Joules	J
Power (P)	Watts	W
Resistance (R)	Ohms	Ω
Voltage / Potential difference (V)	Volts	V

Table 10.1: Units used in **electric circuits**

Exercise 10 – 3:

1. [IEB 2001/11 HG1] - **Emf**
   a) Explain the meaning of each of these two statements:
      i. "The current through the battery is 50 mA."
      ii. "The emf of the battery is 6 V."
   b) A battery tester measures the current supplied when the battery is connected to a resistor of 100 Ω. If the current is less than 50 mA, the battery is "flat" (it needs to be replaced). Calculate the maximum internal resistance of a 6 V battery that will pass the test.
2. [IEB 2005/11 HG] The electric circuit of a torch consists of a cell, a switch and a small light bulb, as shown in the diagram below.



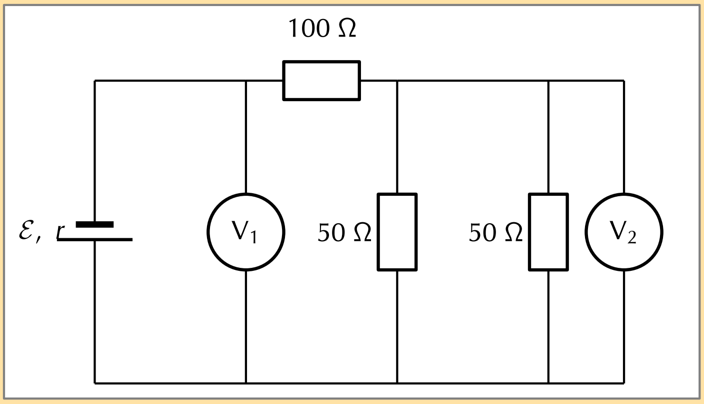
The electric torch is designed to use a D-type cell, but the only cell that is available for use is an AA-type cell. The specifications of these two types of cells are shown in the table below:

Cell	emf	Appliance for which it is designed	Current drawn from cell when connected to the appliance for which it is designed
D	1,5 V	torch	300 mA
AA	1,5 V	TV remote control	30 mA

What is likely to happen and why does it happen when the AA-type cell replaces the D-type cell in the electric torch circuit?

	What happens	Why it happens
(a)	the bulb is dimmer	the AA-type cell has greater internal resistance
(b)	the bulb is dimmer	the AA-type cell has less internal resistance
(c)	the brightness of the bulb is the same	the AA-type cell has the same internal resistance
(d)	the bulb is brighter	the AA-type cell has less internal resistance

3. [IEB 2005/11 HG1] A battery of emf ε and internal resistance r = 25 Ω is connected to this arrangement of resistors.



The resistances of voltmeters V_1 and V_2 are so high that they do not affect the current in the circuit.

a) Explain what is meant by "the emf of a battery".
   The power dissipated in the 100 Ω resistor is 0,81 W.
b) Calculate the current in the 100 Ω resistor.
c) Calculate the reading on voltmeter V_2.
d) Calculate the reading on voltmeter V_1.
e) Calculate the emf of the battery.

4. [SC 2003/11] A kettle is marked 240 V; 1500 W.

a) Calculate the resistance of the kettle when operating according to the above specifications.

b) If the kettle takes 3 minutes to boil some water, calculate the amount of electrical energy transferred to the kettle.

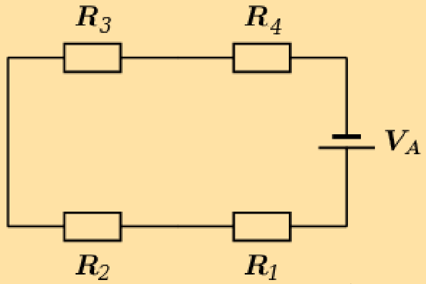
5. [IEB 2001/11 HG1] - **Electric Eels**

   Electric eels have a series of cells from head to tail. When the cells are activated by a nerve impulse, a potential difference is created from head to tail. A healthy electric eel can produce a potential difference of 600 V.

   a) What is meant by "a potential difference of 600 V"?

   b) How much energy is transferred when an electron is moved through a potential difference of 600 V?

6. The diagram shows an electric circuit consisting of a battery and four resistors.



   The **potential difference** (voltage) over the battery is $V_A = 7{,}6$ V

   The the resistors are rated as follows:

   - $R_1 = 4{,}7\ \Omega$
   - $R_2 = 6{,}9\ \Omega$
   - $R_3 = 4{,}9\ \Omega$
   - $R_4 = 4{,}3\ \Omega$

   Assume that positive charge is flowing in the circuit (conventional current).

   Using the concepts of **Ohm's law**, and **electric circuits**, determine the following:

   a) What type of circuit is shown in the diagram?

   b) What is the **total equivalent resistance** R_{eq} of the circuit?

      - round your answer to 1 digit after the decimal comma
      - use the values for any physical constants you might need, as listed here: http://www.everythingscience.co.za/physical-constants

   c) **Question 3**

      What is the potential difference (voltage) across R_1, or V_1 ?

      - round your answer to 3 digits after the decimal comma
      - use the values for any physical constants you might need, as listed here: http://www.everythingscience.co.za/physical-constants

   d) What is the potential difference (voltage) across R_2, R_3, and R_4, or V_2, V_3, and V_4?

      - round your answers to 3 digits after the decimal comma
      - use the values for any physical constants you might need, as listed here: http://www.everythingscience.co.za/physical-constants

7. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27XX 2. 27XY 3. 27XZ 4. 27Y2 5. 27Y3 6. 27Y4

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CHAPTER 11

Electrodynamics

11.1 Introduction ESCQ3

In Grade 11 you learnt how a magnetic field is generated around a current-carrying conductor. You also learnt how a current is generated in a conductor that moves in a magnetic field or in a stationay conductor in a changing magnetic field. This chapter describes how conductors moving in a magnetic field are applied in the real-world.

Today, currents induced by magnetic fields are essential to our technological society. The ubiquitous generator—found in automobiles, on bicycles, in nuclear power plants, and so on—uses magnetism to generate current. Other devices that use magnetism to induce currents include pickup coils in electric guitars, transformers of every size, certain microphones, airport security gates, and damping mechanisms on sensitive chemical balances. Not so familiar perhaps, but important nevertheless, is that the behavior of AC circuits depends strongly on the effect of magnetic fields on currents.

See video: 27Y5 at www.everythingscience.co.za

11.2 Electrical machines - generators and motors ESCQ4

We have seen that when a conductor is moved in a magnetic field or when a magnet is moved near a conductor, a current flows in the conductor. The amount of current depends on:

- the speed at which the conductor experiences a changing magnetic field,
- the number of coils that make up the conductor, and
- the position of the plane of the conductor with respect to the magnetic field.

The effect of the orientation of the conductor with respect to the magnetic field is illustrated in Figure 11.1.

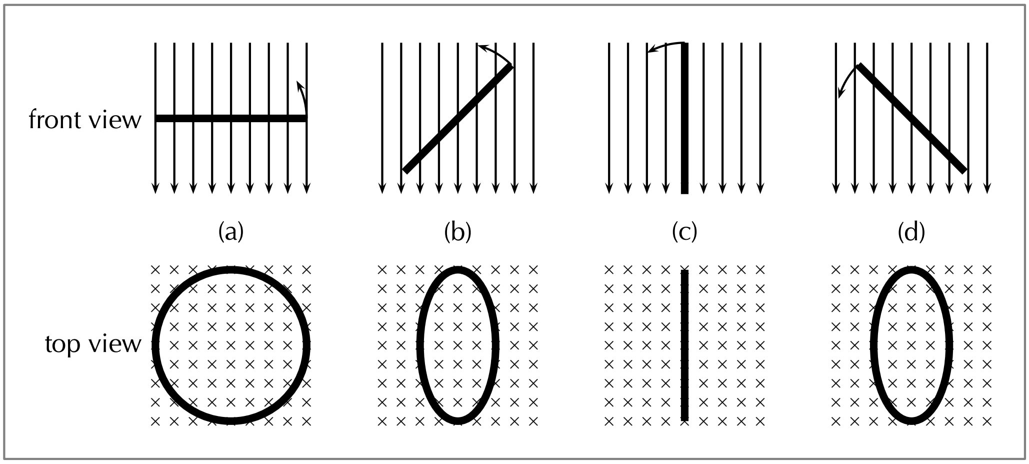


Figure 11.1: Series of figures showing that the magnetic flux through a conductor is dependent on the angle that the plane of the conductor makes with the magnetic field. The greatest flux passes through the conductor when the plane of the conductor is perpendicular to the magnetic field lines as in Figure 11.1 (a). The number of field lines passing through the conductor decreases, as the conductor rotates until it is parallel to the magnetic field Figure 11.1 (c).

If the emf induced and the current in the conductor were plotted as a function of the angle between the plane of the conductor and the magnetic field for a conductor that has a constant speed of rotation, then the induced emf and current would vary as shown in 11.2. The current alternates around zero and is known as an *alternating current* (abbreviated AC).

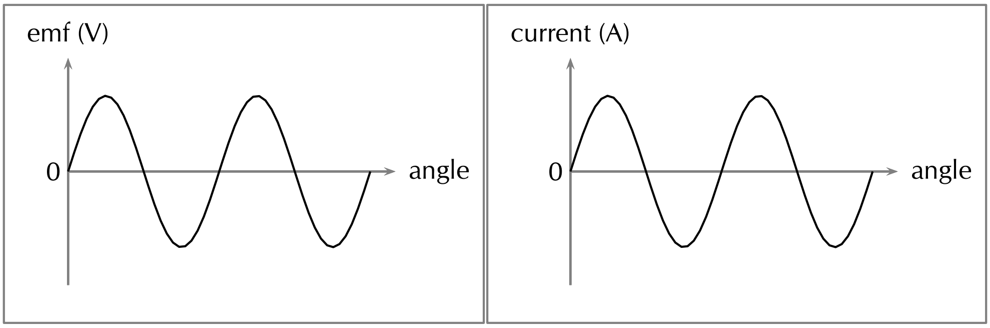


Figure 11.2: Variation of induced emf and current as the angle between the plane of a conductor and the magnetic field changes.

The angle changes as a function of time so the above plots can be mapped onto the time axis as well.

Recall Faraday's Law, which you learnt about in Grade 11:

DEFINITION: *Faraday's Law*

The emf, $\mathcal{E}$, induced around a single loop of conductor is proportional to the rate of change of the magnetic flux, ϕ, through the area, A, of the loop. This can be stated mathematically as:

$$\mathcal{E} = -N\frac{\Delta\phi}{\Delta t}$$

where $\phi = B \cdot A\cos\theta$ and B is the strength of the magnetic field.

Faraday's Law relates induced emf to the rate of change of magnetic flux, which is the product of the magnetic field strength and the cross-sectional area the field lines pass through. The cross-sectional area changes as the loop of the conductor rotates which gives rise the $\cos\theta$ factor. θ is the angle between the normal to the surface area of the loop of the conductor and the magnetic field. As the closed loop conductor changes orientation with respect to the magnetic field, the amount of magnetic flux through the area of the loop changes and an emf is induced in the conducting loop.

Electrical generators ESCQ5

AC generator ESCQ6

The principle of rotating a conductor in a magnetic field to generate current is used in electrical generators. A generator converts mechanical energy (motion) into electrical energy.

DEFINITION: *Generator*

A generator is a device that converts mechanical energy into electrical energy.

FACT

AC generators are also known as alternators. They are found in motor cars to charge the car battery.

The layout of a simple AC generator is shown in Figure 11.3. The conductor is formed of a coil of wire, placed inside a magnetic field. The conductor is manually rotated within the magnetic field. This generates an alternating emf. The alternating current needs to be transmitted from the conductor to the load, which is the system requiring the electrical energy to function.

The load and the conductor are connected by a slip ring. A slip ring is a connector which is able to transmit electricity between rotating portions of a machine. It is made up of a ring and brushes, one of which is stationary with respect to the other. Here, the ring attaches to the conductor and the brushes are attached to the load. Current is generated in the rotating conductor, passes into the slip rings, which rotate against the brushes. The current is transmitted through the brushes into the load, and the system is thus powered.

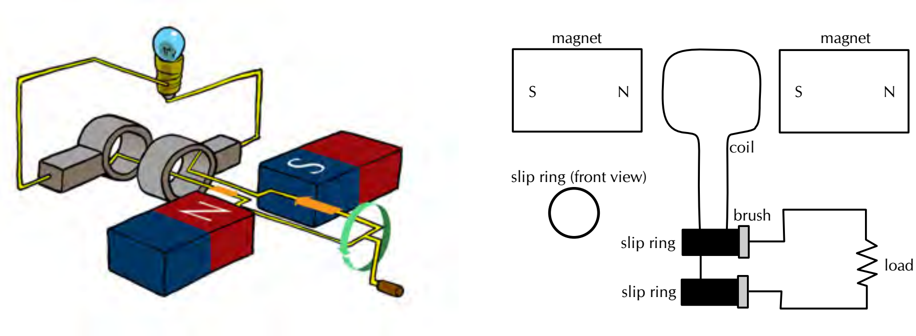


Figure 11.3: Layout of an alternating current generator.

The direction of the current changes with every half turn of the coil. As one side of the loop moves to the other pole of the magnetic field, the current in the loop changes direction. This type of current which changes direction is known as alternating current and Figure 11.4 shows how it comes about as the conductor rotates.

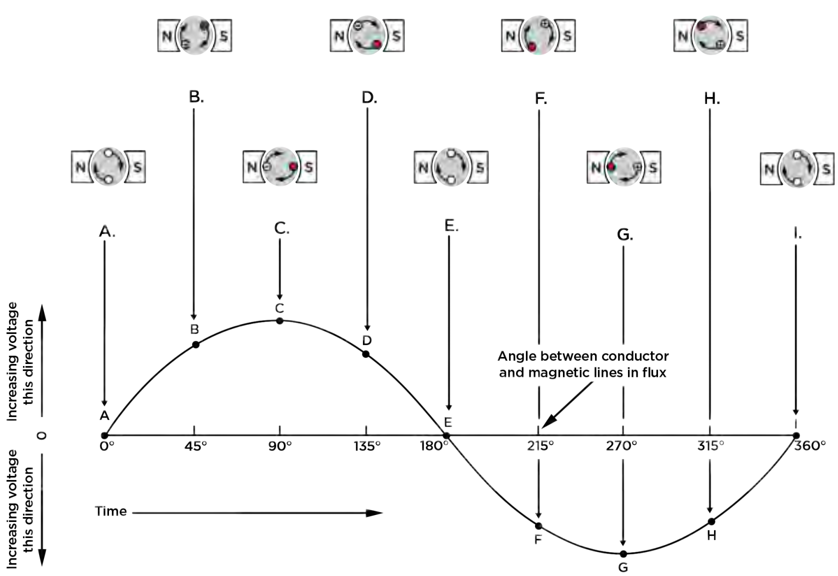


Figure 11.4: The red (solid) dots represent current coming out of the page and the crosses show current going into the page.

DC generator ESCQ7

A simple DC generator is constructed the same way as an AC generator except that there is one slip ring which is split into two pieces, called a commutator, so the current in the external circuit does not change direction. The layout of a DC generator is shown in Figure 11.5. The split-ring commutator accommodates for the change in direction of the current in the loop, thus creating direct current (DC) current going through the brushes and out to the circuit. The current in the loop does reverse direction but if you look carefully at the 2D image you will see that the section of the split-ring commutator also changes which side of the circuit it is touching. If the current changes direction at the same time that the commutator swaps sides the external circuit will always have current going in the same direction.

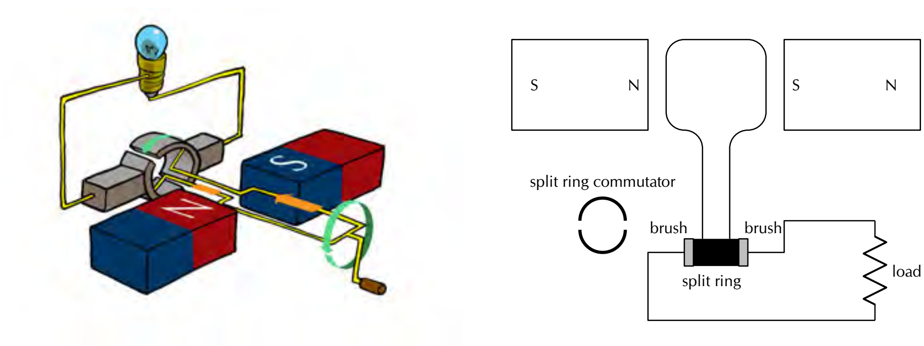


Figure 11.5: Layout of a direct current generator.

The shape of the emf from a DC generator is shown in Figure 11.6. The emf is not steady but is the absolute value of a sine/cosine wave.

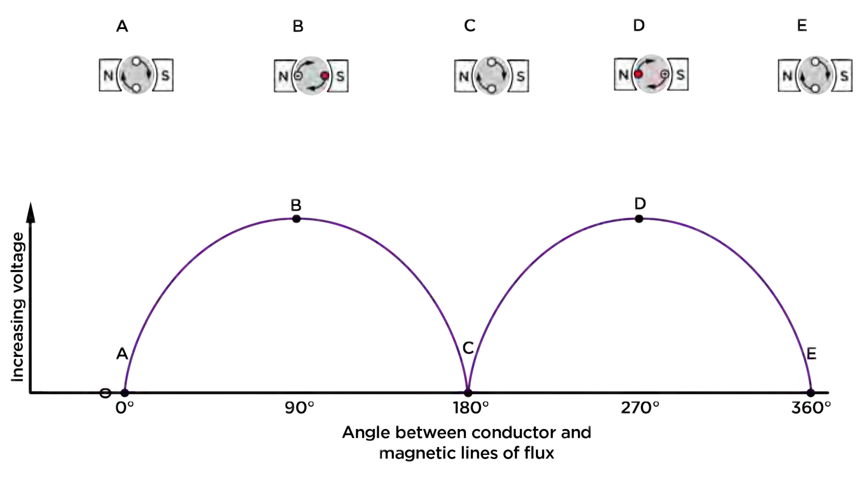


Figure 11.6: Variation of emf in a DC generator.

AC versus DC generators ESCQ8

The problems involved with making and breaking electrical contact with a moving coil are sparking and heat, especially if the generator is turning at high speed. If the atmosphere surrounding the machine contains flammable or explosive vapours, the practical problems of spark-producing brush contacts are even greater.

If the magnetic field, rather than the coil/conductor is rotated, then brushes are not needed in an AC generator (alternator), so an alternator will not have the same problems as DC generators. The same benefits of AC over DC for generator design also apply to electric motors. While DC motors need brushes to make electrical contact with moving coils of wire, AC motors do not. In fact, AC and DC motor designs are very similar to their generator counterparts. The AC motor is depends on the reversing magnetic field produced by alternating current through its stationary coils of wire to make the magnet rotate. The DC motor depends on the brush contacts making and breaking connections to reverse current through the rotating coil every 1/2 rotation (180 degrees).

Electric motors ESCQ9

The basic principles of operation for an electric motor are the same as that of a generator, except that a motor converts electrical energy into mechanical energy (motion).

DEFINITION: *Electric motor*

An electric motor is a device that converts electrical energy into mechanical energy.

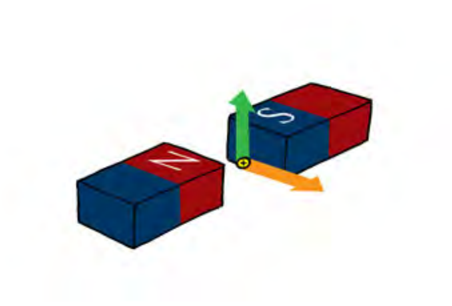
If one were to place a moving charged particle in a magnetic field, it would experience a force called the **Lorentz force**.

DEFINITION: *The Lorentz Force*

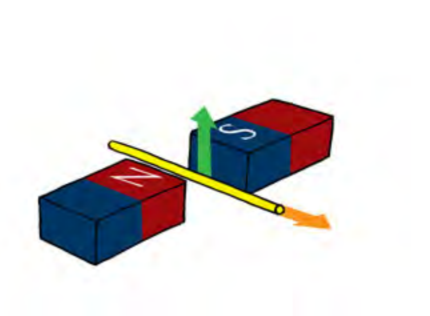
The Lorentz force is the force experienced by a moving charged particle in an electric and magnetic field. The magnetic component is:

$$F = qvB$$

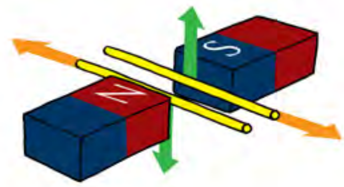
where F is the force (in newtons, N), q is the electric charge (in coulombs, C), v is the velocity of the charged particle (in $m \cdot s^{-1}$) and B is the magnetic field strength (in teslas, T).

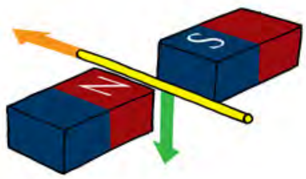


In this diagram a positive charge is shown moving between two opposite poles of magnets. The direction of the charge's motion is indicated by the orange arrow. It will experience a Lorentz force which will be in the direction of the green arrow.

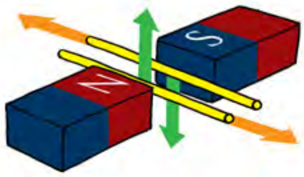


A current-carrying conductor, where the current is in the direction of the orange arrow, will also experience a magnetic force, the green arrow, due to the Lorentz force on the individual charges moving in the current flow.

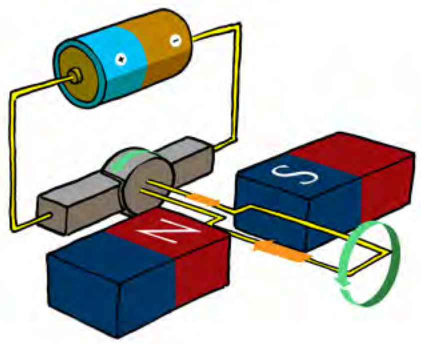




If the direction of the current is reversed, for the same magentic field direction, then the direction of the magnetic force will also be reversed as indiced in this diagram.

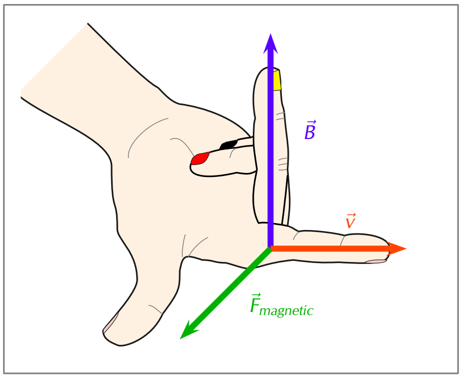


We can see that if there are two parallel conductors with current in opposite direcions they will experience magnetic forces in opposite directions.



An electric motor works by using a source of emf to make a current flow in a loop of conductor such that the Lorentz force on opposite sides of the loop are in opposite directions which can cause the loop to rotate about a central axis.

The direction of the magnetic force is perpendicular to both the direction of the flow of current and the direction of the magnetic field and can be found using the **Right Hand Rule** as shown in the picture below. Use your *right hand*; your first finger points in the direction of the current, your second finger in the direction of the magnetic field and your thumb will then point in the direction of the force.



Both motors and generators can be explained in terms of a coil that rotates in a magnetic field. In a generator the coil is attached to an external circuit that is turned, resulting in a changing flux that induces an emf. In a motor, a current-carrying coil in a magnetic field experiences a force on both sides of the coil, creating a twisting force (called a *torque*, pronounce like 'talk') which makes it turn.

If the current is AC, the two slip rings are required to create an AC motor. An AC motor is shown in 11.7

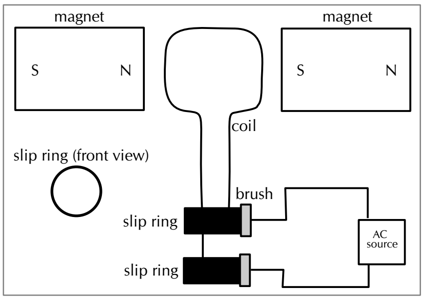


Figure 11.7: Layout of an AC motor.

If the current is DC, split-ring commutators are required to create a DC motor. This is shown in 11.8.

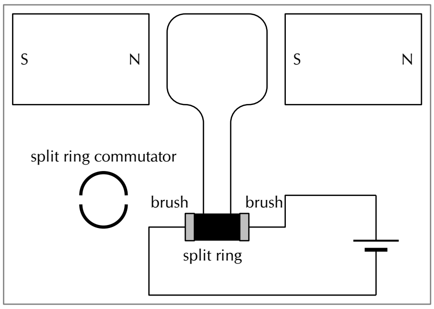


Figure 11.8: Layout of a direct current motor.

Real-life applications ESCQB

Cars

A car contains an alternator. When the car's engine is running the alternator charges its battery and powers the car's electric system.

Project: Alternators

Try to find out the different current values produced by alternators for different types of machines. Compare these to understand what numbers make sense in the real world. You will find different values for cars, trucks, buses, boats etc. Try to find out what other machines might have alternators.

A car also contains a DC electric motor, the starter motor, to turn over the engine to start it. A starter motor consists of the very powerful DC electric motor and starter solenoid that is attached to the motor. A starter motor requires a very high current to crank the engine and is connected to the battery with large cables to carry large current.

Electricity generation

In order to produce electricity for mass distribution (to homes, offices, factories and so forth), AC generators are usually used. The electricity produced by massive power plants usually has a low voltage which is converted to high voltage. It is more efficient to distribute electricity over long distances in the form of high voltage power lines.

The high voltages are then coverted to 240 V for consumption in homes and offices. This is usually done within a few kilometres of where it will be used.

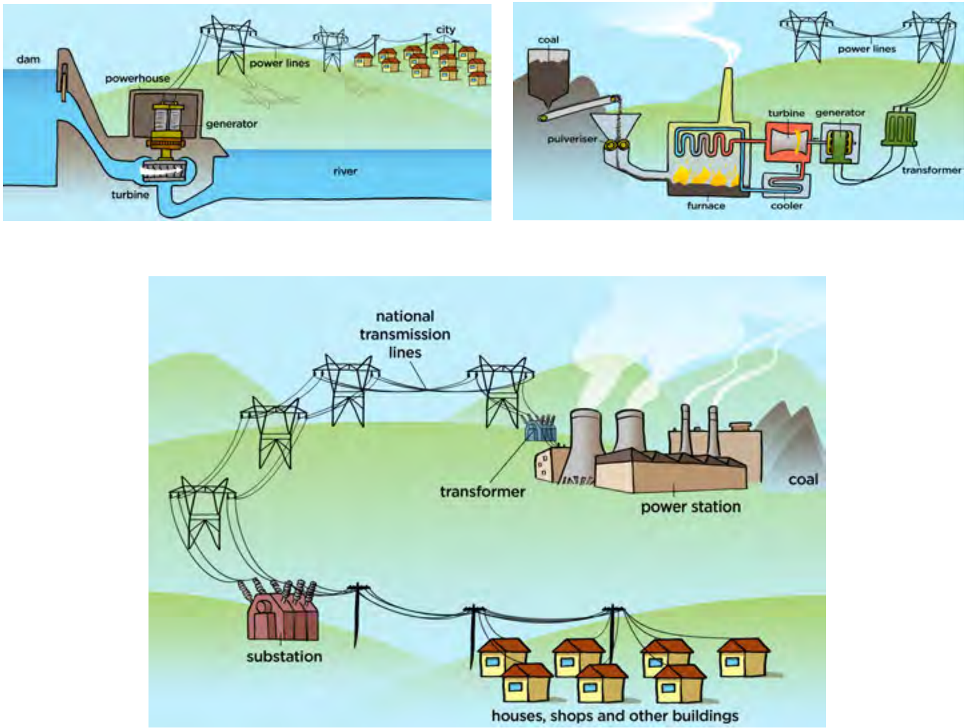


Figure 11.9: AC generators are used at power plants (all types, hydro- and coal-plants shown in top row) to generate electricity.

Exercise 11 – 1: Generators and motors

1. State the difference between a generator and a motor.
2. Use Faraday's Law to explain why a current is induced in a coil that is rotated in a magnetic field.
3. Explain the basic principle of an AC generator in which a coil is mechanically rotated in a magnetic field. Draw a diagram to support your answer.
4. Explain how a DC generator works. Draw a diagram to support your answer. Also, describe how a DC generator differs from an AC generator.
5. Explain why a current-carrying coil placed in a magnetic field (but not parallel to the field) will turn. Refer to the force exerted on moving charges by a magnetic field and the torque on the coil.
6. Explain the basic principle of an electric motor. Draw a diagram to support your answer.
7. Give examples of the use of AC and DC generators.
8. Give examples of the uses of motors.
9. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27Y6 2. 27Y7 3. 27Y8 4. 27Y9 5. 27YB 6. 27YC
7. 27YD 8. 27YF

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FACT
In South Africa alternating current is generated at a frequency of 50 Hz.

11.3 Alternating current ESCQC

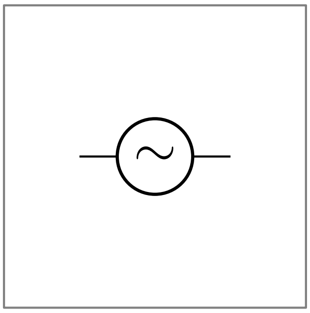
Most students learning about electricity begin with what is known as *direct current* (DC), which is electricity flowing in one direction only. DC is the kind of electricity made by a battery, with definite positive and negative terminals.

However, we have seen that the electricity produced by some generators constantly alternates (switches direction) and is therefore known as *alternating current* (AC). There are a number of advantages to AC current, the main advantage to AC is that the voltage can be changed using transformers. That means that the voltage can be "stepped up" at power stations to a very high voltage so that electrical energy can be transmitted along power lines at low current and therefore experience low energy loss due to heating. The voltage can then be stepped down for use in buildings and street lights.

A list of the advantages of AC current:

- Easy to be transformed (step up or step down using a transformer).
- Easier to convert from AC to DC than from DC to AC.
- Easier to generate.
- It can be transmitted at high voltage and low current over long distances with less energy lost.
- High frequency used in AC makes it suitable for motors.

The circuit symbol for alternating current is:



Current and voltage ESCQD

In an ideal DC circuit, current and voltage are constant. In an AC circuit, current and voltage vary with time. The value of the current or voltage at any specific time is called the *instantaneous current or voltage* and is calculated as follows:

$$i = I_{\max} \sin(\omega t)$$
$$v = V_{\max} \sin(\omega t)$$

i is the instantaneous current. $I_{\max}$ is the maximum current. v is the instantaneous voltage. $V_{\max}$ is the maximum voltage. f is the frequency of the AC and t is the time at which the instantaneous current or voltage is being calculated.

The value we use for AC is known as the root mean square (rms) average. This is the same as what the DC voltage would be for the same source and is defined as:

$$I_{rms} = \frac{I_{\max}}{\sqrt{2}}$$
$$V_{rms} = \frac{V_{\max}}{\sqrt{2}}$$

Since AC varies sinusoidally, with as much positive as negative, doing a straight average would get you zero for the average voltage. The rms value by-passes this problem.

Worked example 1: Laptop transformer

QUESTION

The transformer for the laptop on which this book was written has the following information:

- INPUT: 100-240 V; 1,5 A; 50/60 Hz
- OUTPUT:20 V; 3,25 A

What changes from input to output, apart from the voltage and current values, and what does that imply? In addition, calculate the rms (root mean square) current and voltage values for the input and/or output as appropriate.

SOLUTION

Step 1: Comparing input and output

The input description includes a frequency because it is designed for regular household use where we use alternating current. The output doesn't include a frequency. This implies that the output is not alternating current. This means that the output voltage and current will be constant with time.

Step 2: RMS values

Root mean square values are only applicable when dealing with alternating current. The transformer takes alternating current input and produces direct current output, this means that we only need to determine the rms values for the input.

$$\begin{aligned} V_{\text{rms}} &= \frac{V_{\max}}{\sqrt{2}} \\ V_{\text{rms}} &= \frac{240\text{ V}}{\sqrt{2}} \\ &= 169{,}71\text{ V} \end{aligned}$$

$$\begin{aligned} I_{\text{rms}} &= \frac{I_{\max}}{\sqrt{2}} \\ I_{\text{rms}} &= \frac{1{,}5\text{ A}}{\sqrt{2}} \\ &= 1{,}06\text{ A} \end{aligned}$$

Therefore $V_{\text{rms}} = 169{,}71$ V Therefore $I_{\text{rms}} = 1{,}06$ A

Worked example 2: Camera battery charger

QUESTION

A camera charger has the following information:

- INPUT: 100-240 V; 0,085 A (100 V) - 0,05 A (240 V) ; 50/60 Hz
- OUTPUT:4,2 V; 0,7 A

Calculate the rms (root mean square) current and voltage values for both 100 V and 240 V input.

SOLUTION

Step 1: Understanding the two cases

The reason the transformer has the different input voltages listed is because it may be used internationally and not all countries use the same household voltage. The transformers purpose is to ensure that the output is consistent regardless of the input voltage. The different input voltages of 100 V and 240 V result in different input current values. This is why two different current values are listed under input but the voltage in parentheses tells you which case they are applicable to.

The cases are:

- **100 V:** 0,085 A
- **240 V:** 0,05 A

Step 2: Input voltage of 100 V

$$\begin{aligned} V_{\text{rms}} &= \frac{V_{\text{max}}}{\sqrt{2}} \\ V_{\text{rms}} &= \frac{100\text{ V}}{\sqrt{2}} \\ &= 70{,}71\text{ V} \end{aligned}$$

$$\begin{aligned} I_{\text{rms}} &= \frac{I_{\text{max}}}{\sqrt{2}} \\ I_{\text{rms}} &= \frac{0{,}085\text{ A}}{\sqrt{2}} \\ &= 0{,}06\text{ A} \end{aligned}$$

Therefore $V_{\text{rms}} = 70{,}71$ V

Therefore $I_{\text{rms}} = 0{,}06$ A

Step 3: Input voltage of 240 V

$$\begin{aligned} V_{\text{rms}} &= \frac{V_{\text{max}}}{\sqrt{2}} \\ V_{\text{rms}} &= \frac{240\text{ V}}{\sqrt{2}} \\ &= 169{,}71\text{ V} \end{aligned}$$

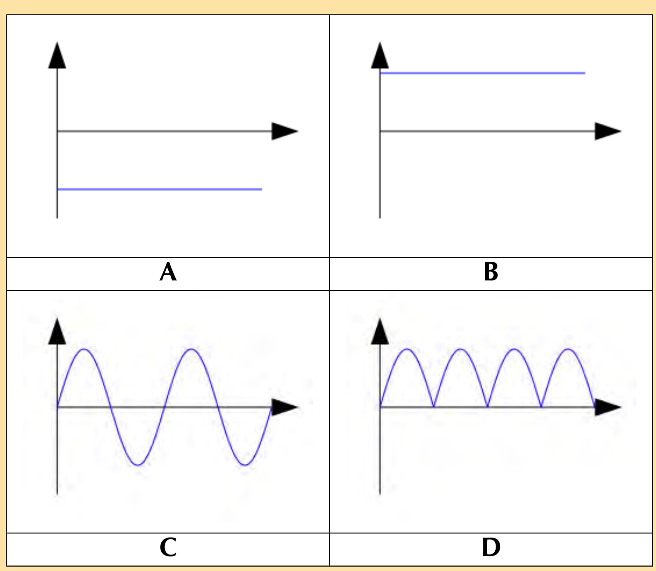
$$\begin{aligned} I_{\text{rms}} &= \frac{I_{\text{max}}}{\sqrt{2}} \\ I_{\text{rms}} &= \frac{0{,}5\text{ A}}{\sqrt{2}} \\ &= 0{,}35\text{ A} \end{aligned}$$

Therefore $V_{\text{rms}} = 169{,}71$ V

Therefore $I_{\text{rms}} = 0{,}35$ A

Exercise 11 – 2: Alternating current

1. Explain the advantages of alternating current.
2. Which of the following graphs correctly shows the current vs. time graph for an AC generator?



3. Write expressions for the current and voltage in an AC circuit.
4. Define the rms (root mean square) values for current and voltage for AC.
5. What is the frequency of the AC generated in South Africa?
6. If $V_{\max}$ at a power station generator is 340 V AC, what is the mains supply (rms voltage) in our household?
7. Given: $I_{\max}$ is 10 A
   Calculate the rms (root mean square) current to two decimal places.
8. Given: $V_{\max}$ is 266 V
   Calculate the rms (root mean square) voltage to two decimal places.
9. Draw a graph of voltage vs time and current vs time for an AC circuit.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27YG 2. 27YH 3. 27YJ 4. 27YK 5. 27YM 6. 27YN
7. 27YP 8. 27YQ 9. 27YR

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Power ESCQF

If the current and voltage are functions of time, so they are always changing, then so will the power that is dissipated in any circuit element. In circuits which contain only ohmic resistors the **average** power dissipated in any component can be calculated in terms of the rms values.

$$\begin{aligned} P_{av} &= I_{rms}V_{rms} \\ &= \frac{I_{\max}}{\sqrt{2}}\frac{V_{\max}}{\sqrt{2}} \\ &= \frac{1}{2}I_{\max}V_{\max} \end{aligned}$$

You might ask why we don't need to use an rms value for power. In an AC circuit both the current and voltage have the same sign so they are both either positive or negative. This means that power, the product of the two is always positive. If power is always positive then the average value won't be zero as in the case of current or voltage in AC circuits.

Worked example 3: Laptop transformer power

QUESTION

The transformer for a laptop has the following information:

- INPUT: 100-240 V; 1,5 A; 50/60 Hz
- OUTPUT:20 V; 3,25 A

Using the input values and assuming 240 V, what is the average power dissipated in the transformer?

SOLUTION

Step 1: RMS values

As calculated previously:

$$V_{\text{rms}} = \frac{V_{\text{max}}}{\sqrt{2}}$$
$$V_{\text{rms}} = \frac{240\text{ V}}{\sqrt{2}} = 169{,}71\text{ V}$$

$$I_{\text{rms}} = \frac{I_{\text{max}}}{\sqrt{2}}$$
$$I_{\text{rms}} = \frac{1{,}5\text{ A}}{\sqrt{2}} = 1{,}06\text{ A}$$

Therefore $V_{\text{rms}} = 169{,}71$ V

Therefore $I_{\text{rms}} = 1{,}06$ A

Step 2: Average power

$$\begin{aligned} P_{av} &= I_{rms}V_{rms} \\ &= 1{,}06\text{ A} \cdot 169{,}71\text{ V} \\ &= 179{,}89\text{ W} \end{aligned}$$

Worked example 4: Motors and generators [NSC 2011 Paper 1]

QUESTION

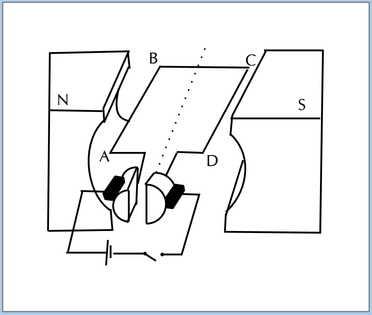
Diesel-electric trains make use of electric motors as well as generators.

1. The table below compares a motor and a generator in terms of the type of energy conversion and the underlying principle on which each operates. Complete the table by writing down only the question number (11.1.1–11.1.4) in the ANSWER BOOK and next to each number the answer.

	TYPE OF ENERGY CONVERSION	PRINCIPLE OF OPERATION
Motor	11.1.1	11.1.3
Generator	11.1.2	11.1.4

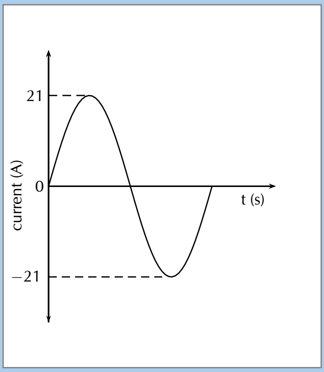
(4 marks)

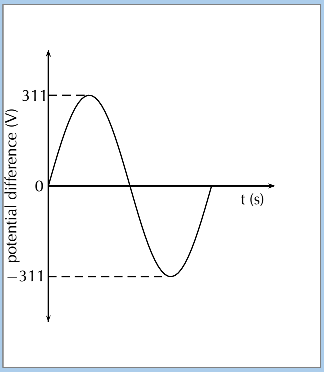
2. The simplified diagram below represents an electric motor.



Give a reason why the section of the coil labelled **BC** in the above diagram does not experience a magnetic force whilst the coil is in the position as shown. (2 marks)

3. Graphs of the current and potential difference outputs of an AC generator are shown below.





Calculate the average power output of this generator. (6 marks)

[TOTAL: 12 marks]

SOLUTION

Question 1

1. Electrical (energy) to mechanical (kinetic) energy
2. Mechanical (kinetic) energy to electrical (energy)
3. Motor effect
4. Electromagnetic induction

(4 marks)

Question 2

BC (conductor) is parallel to the magnetic field.

OR

Open switch, no current (2 marks)

Question 3

Option 1:

$$\begin{aligned} P_{ave} &= V_{rms} I_{rms} \\ &= \frac{V_{max}}{\sqrt{2}} \cdot \frac{I_{max}}{\sqrt{2}} \\ &= \frac{(311)(21)}{2} \\ &= 3265{,}5 \text{ W} \end{aligned}$$

OR

$$\begin{aligned} P_{max} &= V_{max} I_{max} \\ &= (311)(21) \\ &= 6531 \text{ W} \end{aligned}$$

$$\begin{aligned} \therefore P_{ave} &= \frac{P_{max}}{2} \\ &= \frac{6531}{2} \\ &= 3265{,}5 \text{ W} \end{aligned}$$

Option 2:

$$\begin{aligned} V_{rms} &= \frac{V_{max}}{\sqrt{2}} \\ &= \frac{311}{\sqrt{2}} \\ &= 219{,}91 \text{ V} \end{aligned}$$

$$\begin{aligned} I_{rms} &= \frac{I_{max}}{\sqrt{2}} \\ &= \frac{21}{\sqrt{2}} \\ &= 14{,}85 \text{ A} \end{aligned}$$

$$\begin{aligned} P_{ave} &= V_{rms} I_{rms} \\ &= (219{,}91)(14{,}85) \\ &= 3265{,}66 \text{ W} \end{aligned}$$

Option 3:

$$\begin{aligned} R &= \frac{V}{I} \\ &= \frac{311}{21} \\ &= 14{,}81\ \Omega \end{aligned}$$

$$\begin{aligned} I_{\text{rms}} &= \frac{I_{\text{max}}}{\sqrt{2}} \\ &= \frac{21}{\sqrt{2}} \\ &= 14{,}85\ \text{A} \end{aligned}$$

$$\begin{aligned} P_{\text{ave}} &= I_{\text{rms}}^2 R \\ &= (14{,}85)^2(14{,}81) \\ &= 3265{,}83\ \text{W} \end{aligned}$$

Option 4:

$$\begin{aligned} R &= \frac{V}{I} \\ &= \frac{311}{21} \\ &= 14{,}81\ \Omega \end{aligned}$$

$$\begin{aligned} V_{\text{rms}} &= \frac{V_{\text{max}}}{\sqrt{2}} \\ &= \frac{311}{\sqrt{2}} \\ &= 219{,}91\ \text{V} \end{aligned}$$

$$\begin{aligned} P_{\text{ave}} &= \frac{V_{\text{rms}}^2}{R} \\ &= \frac{(219{,}41)^2}{(14{,}81)} \\ &= 3265{,}83\ \text{W} \end{aligned}$$

(6 marks)

[TOTAL: 12 marks]

11.4 Chapter summary ESCQG

See presentation: 27YS at www.everythingscience.co.za

- Electrical generators convert mechanical energy into electrical energy.
- Electric motors convert electrical energy into mechanical energy.
- There are two types of generators - AC and DC. An AC generator is also called an alternator.
- There are two types of motors - AC and DC.
- Alternating current (AC) has many advantages over direct current (DC) listed previously.
- The root mean square (rms) value of a quantity is the maximum value the quantity can have divided by $\sqrt{2}$.
- RMS values are used for voltage and current when dealing with alternating current, $I_{\text{rms}} = \frac{I_{\text{max}}}{\sqrt{2}}$ and $V_{\text{rms}} = \frac{V_{\text{max}}}{\sqrt{2}}$.
- The average power dissipated in a purely resistive circuit with alternating current is $P_{av} = I_{rms} V_{rms}$.

Exercise 11 – 3:

1. [SC 2003/11] Explain the difference between alternating current (AC) and direct current (DC).
2. Explain how an AC generator works. You may use sketches to support your answer.
3. What are the advantages of using an AC motor rather than a DC motor.
4. Explain how a DC motor works.
5. At what frequency is AC generated by Eskom in South Africa?
6. (IEB 2001/11 HG1) - **Work, Energy and Power in Electric Circuits**

   Mr. Smith read through the agreement with Eskom (the electricity provider). He found out that alternating current is supplied to his house at a frequency of 50 Hz. He then consulted a book on electric current, and discovered that alternating current moves to and fro in the conductor. So he refused to pay his Eskom bill on the grounds that every electron that entered his house would leave his house again, so therefore Eskom had supplied him with nothing!

   Was Mr. Smith correct? Or has he misunderstood something about what he is paying for? Explain your answer briefly.
7. You are building a laser that takes alternating current and it requires a very high peak voltage of 180 kV. By your calculations the entire laser setup can be treated at a single resistor with an equivalent resistance of 795 ohms. What is the rms value for the voltage and the current and what is the average power that your laser is dissipating?

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27YT 2. 27YV 3. 27YW 4. 27YX 5. 27YY 6. 27YZ
7. 27Z2

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